# COMPUTING SMALLEST CONVEX INTERSECTING POLYGONS*

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Abstract. A polygon $C$ is an *intersecting polygon* for a set $\mathcal{O}$ of objects in $\mathbb{R}^2$ if $C$ intersects each object in $\mathcal{O}$, where the polygon includes its interior. We study the problem of computing the minimum-perimeter intersecting polygon and the minimum-area convex intersecting polygon for a given set $\mathcal{O}$ of objects. We present an FPTAS for both problems for the case where $\mathcal{O}$ is a set of possibly intersecting convex polygons in the plane of total complexity $n$.

Furthermore, we present an exact polynomial-time algorithm for the minimum-perimeter intersecting polygon for the case where $\mathcal{O}$ is a set of $n$ possibly intersecting segments in the plane. So far, polynomial-time exact algorithms were only known for the minimum-perimeter intersecting polygon of lines or of disjoint segments.

## 1 Introduction

Convex hulls are among the most fundamental objects studied in computational geometry. In fact, the problem of designing efficient algorithms to compute the convex hull of a planar point set $\mathcal{O}$—the smallest convex set containing $\mathcal{O}$—is one of the problems that started the field [10, 19]. Since the early days, the problem has been studied extensively, resulting in practical and provably efficient algorithms, in the plane as well as in higher dimensions; see the survey by Seidel [21, Chapter 26] for an overview.

A natural generalization is to consider convex hulls for a collection $\mathcal{O}$ of geometric objects (instead of points) in $\mathbb{R}^2$. Note that the convex hull of a set of polygonal objects is the same as the convex hull of the vertices of the objects. Hence, such convex hulls can be computed using algorithms for computing the convex hull of a point set. A different generalization, which leads to more challenging algorithmic questions, is to consider the smallest convex set that *intersects* all objects in $\mathcal{O}$. Thus, instead of requiring the convex set to fully contain each object from $\mathcal{O}$, we only require that it has a non-empty intersection with each object.

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Notice that in case of points, the "smallest" set with respect to containment is well-defined: if convex sets $C_1$ and $C_2$ both contain a point set $\mathcal{O}$, then $C_1 \cap C_2$ also contains $\mathcal{O}$. Hence, the convex hull of a point set $\mathcal{O}$ can be defined as the intersection of all convex sets containing $\mathcal{O}$. When $\mathcal{O}$ consists of objects, however, this is no longer true, and the term "smallest" is ambiguous. In the present paper we consider two variants: given a set $\mathcal{O}$ of possibly intersecting convex polygons in $\mathbb{R}^2$ of total complexity $n$, find a convex set of minimum-perimeter that intersects all objects in $\mathcal{O}$, or a convex set of minimum area that intersects all objects in $\mathcal{O}$.

Observe that a minimum-perimeter connected intersecting set $C$ for $\mathcal{O}$ must be a convex polygon. To see this, observe that for any object $o \in \mathcal{O}$ we can select a point $p_o \in o \cap C$, and take the convex hull of these points; the result is a feasible convex polygon whose perimeter is no longer than that of $C$. Thus the convexity of the solution could be omitted from the problem statement. This contrasts with the minimum-area problem, where there is always an intersecting polygon of zero area, namely a polygonal chain (which is an "infinitely thin" polygon). The convexity requirement is therefore essential in the problem statement. Note that it is still true that the minimum-area convex intersecting set is a polygon: given a convex solution, we can again take the convex hull of the points $p_o$ and get a feasible solution whose area is not greater than the area of the initial convex solution. We also remark that the two problems typically have different optima. If $\mathcal{O}$ consists of the three edges of an equilateral triangle, then the minimum-area solution is a line segment (that is, a degenerate polygon of zero area), whereas the minimum-perimeter solution is the triangle whose vertices are the midpoints of the edges.

The problems of computing minimum-area or minimum-perimeter convex intersecting polygons, as well as several related problems, have already been studied. Dumitrescu and Jiang [8] considered the minimum-perimeter intersecting polygon problem. They gave a constant-factor approximation algorithm as well as a PTAS for the case when the objects in $\mathcal{O}$ are segments or convex polygons. They achieved a running time of $n^{O(1)}/\varepsilon + 2^{O(1/\varepsilon^{2/3})}n$. They also prove that computing a minimum-perimeter intersecting polygon for a set $\mathcal{O}$ of non-convex polygons (or polygonal chains) is NP-hard. For convex input objects, however, the hardness proof fails. Hence, Dumitrescu and Jiang ask the following question.

**Question 1.** Is the problem of computing a minimum-perimeter intersecting polygon of a set of segments NP-hard?

In case of *disjoint* segments, a minimum-perimeter intersecting polygon can be found in polynomial time [11, 12], but for intersecting segments the question is still open.

The problem of computing smallest intersecting polygons for a set $\mathcal{O}$ of objects has also been studied in works on *imprecise points*. Now the input is a set of points, but the the exact locations of the points are unknown. Instead, for each point one is given a region where the point can lie. One can then ask questions such as: what is the largest possible convex hull of the imprecise points? And what is the smallest possible convex hull? If we consider the objects in our input set $\mathcal{O}$ as the regions for the imprecise points, then the latter question is exactly the same as our problem of finding smallest intersecting convex sets. In this setup both the minimum-perimeter and minimum-area problem have been considered, for

sets $\mathcal{O}$ consisting of convex regions of total complexity $n$. There are exact polynomial-time algorithms for minimum (and maximum) perimeter and area, for the special case where $\mathcal{O}$ consists of horizontal line segments or axis-parallel squares [17]. Surprisingly, some of these problems are NP-hard, such as the *maximum*-area/perimeter problems for segments. This gave rise to the study of approximation algorithms and approximation schemes for these problems [16].

In some cases, the minimum-perimeter problem can be phrased as a travelling salesman problem with neighborhoods (TSPN). Here the goal is to find the shortest closed curve intersecting all objects from the given set $\mathcal{O}$. In general, an optimal TSPN tour need not be convex, but one can show that in the case of lines or rays, an optimal tour is always convex: if a convex polygon intersects a line (or a ray), then its boundary intersects the line (resp. the ray). Therefore, computing a minimum-perimeter intersecting polygon of lines (or rays) is the same problem as TSPN with line neighborhoods (resp. ray neighborhoods). TSPN of lines in $\mathbb{R}^2$ admits a polynomial-time algorithm [6]. In higher dimensions, TSPN has a PTAS for hyperplane neighborhoods [1], but notice that this is not the natural generalization of the minimum-intersecting polygon problem. Tan [20] proposed an exact algorithm for TSPN of rays in $\mathbb{R}^2$, but there seems to be an error in the argument; see Appendix A. At the time of writing this article, we believe that a polynomial-time algorithm for TSPN of rays is not known, but there is a constant-factor approximation algorithm due to Dumitrescu [7], as well as a polynomial time approximation scheme (PTAS) [8] that finds a $(1+\varepsilon)$-approximate solution in polynomial time for any fixed constant $\varepsilon > 0$.

**Our results.** Before we resolve Question 1, we first develop dynamic-programming-based approximation schemes for the minimum-perimeter and minimum-area convex intersecting polygon problems. The approach used in these approximation algorithms will serve as a stepping stone for our exact algorithm presented later.

Our first algorithm is a fully polynomial time approximation scheme (FPTAS) for the minimum-perimeter problem of arbitrary convex objects of total complexity $n$.

**Theorem 1.** *Let $\mathcal{O}$ be a set of convex polygons of total complexity $n$ in $\mathbb{R}^2$ and let* OPT *be the minimum-perimeter of an intersecting convex polygon for $\mathcal{O}$. For any given $\varepsilon > 0$, we can compute an intersecting polygon for $\mathcal{O}$ whose perimeter is at most $(1+\varepsilon)\cdot$ OPT, in $O(n^{2.373}/\varepsilon + n/\varepsilon^8)$ time.*

This is a vast improvement over the PTAS given by Dumitrescu and Jiang [8], as the dependence on $1/\varepsilon$ is only polynomial in our algorithm compared to exponential in theirs. This algorithm works in a word-RAM model, where input polygons are defined by the coordinates of their vertices, and where each coordinate is a word of at most $b$ bits. On a standard word RAM one often assumes $b = O(\log n)$, but our algorithm is polynomial even if we allow $b = n^{O(1)}$ bits of precision.

We also get a similar approximation scheme for the minimum-area problem, albeit with a slower running time. Here we rely more strongly on the fact that the objects of $\mathcal{O}$ are convex *polygons*, and an extension to (for example) disks is an interesting open question. The minimum-perimeter FPTAS needs to be adapted to the minimum-area setting.

**Theorem 2.** *Let $\mathcal{O}$ be a set of convex polygons in $\mathbb{R}^2$ with $n$ vertices in total, where the coordinates of each vertex are specified using at most $b$ bits, and let* OPT *be the minimum area of an intersecting convex polygon for $\mathcal{O}$. For any given $\varepsilon > 0$, we can compute a convex intersecting polygon for $\mathcal{O}$ whose area is at most $(1+\varepsilon) \cdot$ OPT, in $O(n^{13}\log(b/\varepsilon) + n^{11}/\varepsilon^{24})$ time.*

While the dynamic-programming algorithm developed above is crucial to get an exact algorithm, we are still several steps from being able to resolve Question 1. The main challenge here is that the vertices of the optimum intersecting polygon can be located at arbitrary boundary points in $\mathcal{O}$, and there is no known way to discretize the problem. To overcome this challenge, we introduce a subroutine that uses an algorithm of Dror, Efrat, Lubiw and Mitchell [6] to compute parts of the minimum-perimeter intersecting polygon that contain no vertices of input objects, and show how to combine this with a dynamic-programming approach. This way we are able to achieve a polynomial-time algorithm (on a real-RAM machine) for the minimum-perimeter intersecting polygon problem when the objects are line segments.

**Theorem 3.** *Let $\mathcal{O}$ be a set of $n$ line segments in the plane. Then we can compute a minimum-perimeter intersecting polygon for $\mathcal{O}$ in $O(n^9 \log n)$ time.*

If $\mathtt{P} \neq \mathtt{NP}$, then this gives a direct negative answer to Question 1. The theorem also extends to the case of rays (this is the scenario studied by Tan [20]; see Appendix A).

**Our techniques.** Our approximation algorithms compute an approximate solution whose vertices are from some fine grid. To determine a suitable grid resolution, we need to be able to compute lower bounds on OPT, which is non-trivial. It is also non-trivial to know where to place the grid, such that it is guaranteed to contain an approximate solution. The problem is that our lower bound gives us the location of a solution that is a constant-factor approximation, but this can be far from the location of a $(1+\varepsilon)$-approximation. Hence, for the minimum-area problem we generate a collection of grids, one of which is guaranteed to contain a $(1+\varepsilon)$-approximate solution. Finally, we face some further difficulties since a square grid may be insufficient: the optimum intersecting polygon may be extremely thin and long, and of area close to zero. In such cases there is no square grid of polynomial size that would contain a good solution. These problems are resolved in Section 2.

Section 3 presents our dynamic-programming algorithm (DP, for short) for the minimum-perimeter problem. The main technical difficulty to obtain a DP lies in the fact that it is not clear what subset of objects should be visited in each subproblem. A portion of the optimum's boundary could in principle be tasked with intersecting an arbitrary subset of $\mathcal{O}$, while some of the objects in $\mathcal{O}$ need not be intersected by the optimum boundary and will simply be covered by the interior of the optimum intersecting polygon: a naïve approach therefore would not yield a polynomial-time algorithm. Our carefully designed subproblems have a clear corresponding set of objects to "visit", using orderings of certain tangents of input objects for this purpose. The minimum area problem uses a similar DP, see Section 4 for its details.

Finally, in order to present our exact algorithm in Section 5, we need to modify our DP to deal with subproblems where the vertices of a convex chain do not come from a discretized set. In such cases, we have to find the order in which the objects of $\mathcal{O}$ are visited by the chain. We are able to prove a specific ordering in the case when the objects are line segments. The order then allows us to invoke the algorithm of Dror, Efrat, Lubiw and Mitchell [6] in a black-box manner.

## 2 Locating an Optimal Solution

The algorithms to be presented in subsequent sections need to approximately know the size and location of a smallest intersecting polygon. For the minimum-perimeter intersecting polygon, we will explain that we can use an algorithm by Dumitrescu and Jiang [8] to locate the minimum-perimeter intersecting polygon. For the minimum-area intersecting polygon, we prove that if there does not exist a solution with a constant number of vertices (that can be computed with a different algorithm), then it is sufficient to consider polygons whose vertices are from a grid which comes from a polynomial collection of different grids.

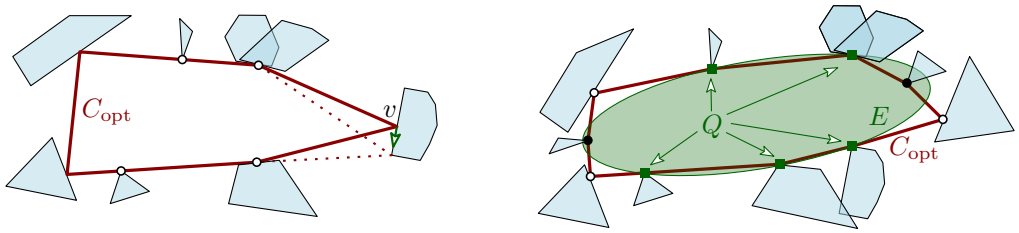
**Locating the minimum-perimeter optimum.** For the minimum-perimeter intersecting polygon of a set $\mathcal{O}$ of convex objects, Dumitrescu and Jiang [8] present an algorithm $A1$ that, for a given $\varepsilon_1 > 0$, outputs a rectangle $R$ intersecting all input objects in $\mathcal{O}$ and with perimeter at most $\frac{4}{\pi}(1+\varepsilon_1)\text{OPT}$. At a high level, $A1$ guesses an orientation of the rectangle among $\lceil \frac{\pi}{4\varepsilon_1} \rceil$ discrete orientations and then uses a linear program to identify the smallest perimeter rectangle of that orientation that intersects $\mathcal{O}$. In [8] it is described how Algorithm $A1$ can be used to locate an optimal solution if the input objects are convex polygons. To this end, given any $\varepsilon > 0$, algorithm A1 is run with $\varepsilon_1 = \frac{\varepsilon}{2+\varepsilon}$, giving a rectangle $R$. Let $\sigma$ be the square that is concentric and parallel to $R$ and has a side length of $3 \cdot \text{per}(R)$. Then the following holds.

**Lemma 4** (Lemma 3 in [8])**.** *Suppose that* $\text{per}(R) \geq (1+\varepsilon) \cdot \text{OPT}$. *Then there is an optimum polygon* $C_{\text{opt}}$ *that is covered by* $\sigma$.

Algorithm $A1$ needs to solve $O(1/\varepsilon)$ many linear programs with $O(n)$ variables and $O(n)$ constraints each. Thus $R$ and $\sigma$ can be found in $O(T_{\text{LP}}(n)/\varepsilon)$ time, where $T_{\text{LP}}(n)$ is the running time of an LP solver with $O(n)$ variables and $O(n)$ constraints. The state-of-the-art LP solver by Jiang, Song, Weinstein, and Zhan [13] achieves a running time of $O(n^{2.373})$.

**The shape and location of the minimum-area optimum.** For the rest of this section, let $\mathcal{O}$ be a set of convex polygons, and let $X$ denote the set of vertices in the planar arrangement induced by $\mathcal{O}$, that is, $X$ consists of the vertices of the polygons in $\mathcal{O}$ plus the intersection points between edges of different polygons.

**Lemma 5.** *Let* $C_{\text{opt}}$ *be a minimum-area intersecting polygon for the input* $\mathcal{O}$. *Then for each vertex* $v$ *of* $C_{\text{opt}}$ *there is some* $o_v \in \mathcal{O}$ *such that* $C_{\text{opt}} \cap o_v = \{v\}$, *and consequently,* $v$ *is on the boundary of some object in* $\mathcal{O}$.




$(i)$ $(ii)$

Figure 1: (i) If $v$ and adjacent sides of $C_{\text{opt}}$ are disjoint from $X$, then we can slide $v$ without increasing the area of $C_{\text{opt}}$. (ii) The points of $Q$ (green) and their circumscribed ellipse $E$.

*Proof.* Let $v$ be a vertex of $C_{\text{opt}}$ and let $\ell$ be a line through $v$ tangent to $C_{\text{opt}}$ such that $\ell \cap C_{\text{opt}} = \{v\}$. Let $\ell^+$ be the half-plane defined by $\ell$ such that $\ell^+ \cap C_{\text{opt}} = C_{\text{opt}}$. Now imagine moving $\ell$ over a distance $\varepsilon$ into the interior of $C_{\text{opt}}$, obtaining a line $\ell(\varepsilon)$. Note that $\ell(\varepsilon)^+ \cap C_{\text{opt}}$ cannot be an intersecting set for $\mathcal{O}$ for any $\varepsilon > 0$, otherwise $C_{\text{opt}}$ would not be optimal. Thus as soon as we move $\ell$, we must loose the property that $\ell^+ \cap C_{\text{opt}}$ is an intersecting set, which implies that there is an object $o \in \mathcal{O}$ such that $o \cap C_{\text{opt}} = \{v\}$. □

**Lemma 6.** *Let $C_{\text{opt}}$ be a minimum-area convex intersecting polygon for the input $\mathcal{O}$ that has the minimum number of vertices, and among such polygons has the maximum number of points from $X$ on its boundary. Then for any vertex $v$ of $C_{\text{opt}}$ that is not in $X$, the relative interior of at least one side of $C_{\text{opt}}$ adjacent to $v$ contains a point of $X$.*

*Proof.* Suppose for a contradiction that $v \notin X$ and that the relative interior of the sides in $C_{\text{opt}}$ adjacent to $v$ are disjoint from $X$. By Lemma 5, vertex $v$ must be on the boundary of an input object, and since $v \notin X$ it is in the relative interior of an edge $e$ of an input object and not on any other edge $e'$; see Fig. 1(i). Then we can move $v$ along $e$ while fixing its neighboring vertices $v_1, v_2$ in $C_{\text{opt}}$, without increasing the area of $C_{\text{opt}}$. This is true because the triangle given by $v, v_1, v_2$ has a fixed side $v_1v_2$ and the corresponding height is non-increasing for at least one of the two possible movement directions along $e$. This movement can be continued until we hit a point in $X$, or the angle of the polygon becomes $\pi$ at $v_1$ or $v_2$. As a result, we end up with a feasible polygon $S$ whose area is no greater than that of $C_{\text{opt}}$, and it has one less vertex or at least one more point of $X$ on its boundary. This contradicts the properties of $C_{\text{opt}}$ stated in the lemma. □

We will now prove the key lemma for Theorem 2. It is common in geometric problems to obtain approximation by considering discrete locations from a sufficiently fine square grid $G$. Interestingly, we are not able to discretize this way, but rather we will generate a collection $\mathcal{G}$ of several rectangular grids with the guarantee that some $G \in \mathcal{G}$ is a good discretization. Each of our rectangular grids is created based on a technique that is reminiscent of a construction from Har-Peled's book [9, Lemma 22.7 and Lemma 23.7].

**Lemma 7.** *For any given set of input polygons $\mathcal{O}$ and $0 < \varepsilon < 1$ there is an intersecting polygon $C$ of area $(1+\varepsilon) \cdot \text{OPT}$ that has at most six vertices, or whose vertices are in a*

*rectangular grid $G$ of size $O(1/\varepsilon^3) \times O(1/\varepsilon^3)$ where $G$ belongs to a collection $\mathcal{G}$ of grids that can be generated in $O(n^{10}/\varepsilon^6)$ time.*

*Proof.* Let $C_{\text{opt}}$ be a minimum-area convex intersecting polygon for the input $\mathcal{O}$ that has the minimum number of vertices, and among such polygons has the maximum number of points from $X$ on its boundary. Notice that if $C_{\text{opt}}$ has area 0, then it is a doubled segment, and thus has only two vertices.

Suppose now that $C_{\text{opt}}$ has at least seven vertices. If $a, b$ are consecutive sides along $\partial C_{\text{opt}}$, the boundary of $C_{\text{opt}}$, then by Lemma 6 either their shared vertex or some point in their relative interior is in $X$. Notice that the non-shared vertices of $a$ and $b$ are not included in this argument. Since $C_{\text{opt}}$ has at least seven sides, it will contain at least three pairs of consecutive sides that are pairwise disjoint. Thus $\partial C_{\text{opt}}$ has at least three points from $X$, no two of which are on the same side of $C_{\text{opt}}$. Let $Q = \partial C_{\text{opt}} \cap X$. Since $\text{area}(C_{\text{opt}}) \neq 0$ and $Q$ contains points on at least three distinct sides of $C_{\text{opt}}$, we have that $Q$ forms a convex polygon of positive area.

Let $E$ be the minimum-area ellipse (of any orientation) covering $Q$. Notice that $E$ is uniquely defined by at most five vertices from $Q$, see Fig. 1(ii). Consequently, we can generate in time $O(|X|^5)$ time a set $\mathcal{B}$ of candidate ellipses that is guaranteed to contain $E$. As a consequence of John's Theorem [14, 2] we can scale $E$ by a factor $1/2$ from the center of $E$ to get an ellipse $E'$ that is contained in the convex hull of $Q$ and therefore also in $C_{\text{opt}}$.

Next, we apply an affine transformation that transforms $E$ into a circle. Note that the transformation changes the area of all polygons by the same multiplicative constant and preserves convexity, therefore the transformation preserves optima and multiplicative approximations of area. We will keep using our previous notations on this transformed instance for the rest of this proof. Without loss of generality assume that $E$ is the unit-radius circle centered at the origin. Consequently, $E'$ is the unit-diameter circle centered at the origin.

The points of $Q$ subdivide $\partial C_{\text{opt}}$ into $2|Q|$ distinct sections: the singletons containing the points of $Q$ and the connected components of $\partial C_{\text{opt}} \setminus Q$. Note that by Lemma 6 a section can contain at most two vertices of $C_{\text{opt}}$. Let $D^*$ be the disk of radius $32/\varepsilon$ centered at the origin. A section is called a *spike* if it contains some vertex of $C_{\text{opt}}$ outside $D^*$. If $v$ is a vertex, then let $S(v)$ denote the convex hull of the section containing $v$. We distinguish two cases.

*Case I: $C_{\text{opt}}$ has a vertex outside $D^*$.*

We will show that in this case there is a $(1+\varepsilon)$-approximate solution that has at most six vertices. For a vertex $v$ of $C_{\text{opt}}$ let $q_v, q'_v \in Q$ be the endpoints of the section of $v$, where we set $q_v = q'_v = v$ if the section of $v$ is $\{v\}$. Recall that $C_{\text{opt}}$ contains $E'$ and it also contains $v$, thus it also contains the convex hull of $E' \cup \{v\}$, see Fig. 3(i). Hence, if $v$ is at distance at least $32/\varepsilon$ from the origin, then the convex hull of $E' \cup \{v\}$ has area at least $16/\varepsilon$. Consequently, if $C_{\text{opt}}$ has a spike, then

$$\text{area}(C_{\text{opt}}) \geqslant 16/\varepsilon. \tag{1}$$

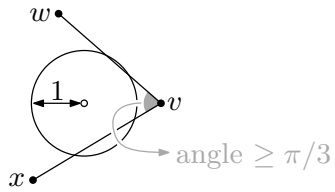



Figure 2: The distance of $v$ to the center of the disk is at most 2.

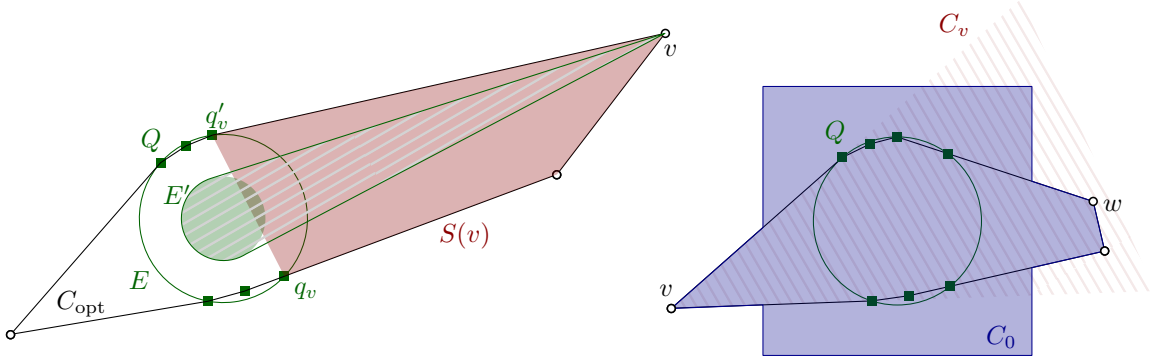



Figure 3: (i) A spike at vertex $v$. The section convex hull $S(v)$ is shaded with red. (ii) Building blocks of a constant-size feasible polygon with at most two spikes and the side-length 4 square. The polygon $C_0$ is shaded blue, and the red falling pattern indicates the cone $C_w$.

*Claim* 8. $C_{\text{opt}}$ has at most two sections that have a vertex at distance strictly greater than 2 from the origin.

*Proof.* Suppose for a contradiction that $v, w, x$ are vertices in disjoint sections, and each of $v, w, x$ are at distance strictly greater than 2 from the origin. It follows that each of the three portions of $\partial C_{\text{opt}}$ defined by $v, w, x$ must contain some point of $Q$, namely, at least one point among the endpoints of the sections containing $v, w$, and $x$. Consequently, each of the three portions of $\partial C_{\text{opt}}$ defined by $v, w, x$ contains some point that falls in the unit disk bounded by the circle $E$. Since the triangle $T$ given by $v, w, x$ is covered by $C_{\text{opt}}$, we have that each side of $T$ must intersect the unit disk (as otherwise at least one of the three boundary portions would be disjoint from the unit disk). Assume without loss of generality that $T$ has its largest angle at $v$. But then $v$ can be at most distance 2 from the origin—this is true since there are two rays emanating from $v$ that intersect the unit disk and make an angle of at least $\pi/3$, see Figure 2—which is a contradiction. ◁

Our claim implies that there are vertices $v, w \in V(C_{\text{opt}})$ such that $C_{\text{opt}}$ can be covered by the union of the section convex hulls $S(v)$, $S(w)$, and the square $[-2, 2] \times [-2, 2]$. Assume without loss of generality that $v$ is a vertex of $C_{\text{opt}}$ at maximum distance from the

origin, and let $w$ be the vertex of $C_{\text{opt}} \setminus S(v)$ at maximum distance from the origin. (We remark that $v$ and $w$ may lie inside $[-2,2] \times [-2,2]$.)

Let $C_0 = S(v) \cup S(w) \cup ([-2,2] \times [-2,2])$. Note that $C_0$ is a (non-convex) polygon covering $C_{\text{opt}}$. Consider the (potentially unbounded) polygon $C_v$ whose sides are the section of $v$, where we extend the sides adjacent to $q_v$ and $q'_v$ beyond $q_v$ and $q'_v$ until they meet (or into rays), see Fig. 3(ii) for an illustration. Notice that $C_v$ covers $C_{\text{opt}}$.

Consider now the polygon $C = C_v \cap C_w$. It is convex and covers $C_{\text{opt}}$, as $C_{\text{opt}}$ is covered by each of $C_v$ and $C_w$ and they are both convex. We claim that $C \subseteq C_0$. Suppose the contrary: there is some point $p \in C \setminus C_0$. Consequently, $p \in C \setminus ([-2,2] \times [-2,2])$ and $p$ is not contained in the unit-radius disk centered at the origin. It follows that $p$ is in some section convex hull $S(x)$ for some vertex $x$ of $C_{\text{opt}}$ such that $S(x)$ is disjoint from $S(v)$ and $S(w)$. Assume without loss of generality that $x$ is the vertex at maximum distance from the origin within $S(x)$. Since $p \in S(x)$ has distance at least 2 from the origin, the same holds for $x$. Recall that by the choice of $v$ and $w$, we know that $v$ and $w$ are at least as far from the origin as $x$. This contradicts Claim 8, as it would mean that each of $S(v), S(w)$, and $S(x)$ have vertices at distance strictly greater than 2 from the origin. This concludes the proof of the claim that $C \subset C_0$. Moreover,

$$\text{area}(C) - \text{area}(C_{\text{opt}}) \leqslant \text{area}([-2,2] \times [-2,2]) = 16.$$

Thus if $C_{\text{opt}}$ has a spike, then by the bound (1) the polygon $C$ has area at most

$$\text{area}(C_{\text{opt}}) + 16 \leqslant (1+\varepsilon)\text{area}(C_{\text{opt}}).$$

Observe that $C$ has at most six vertices: each side of $C$ is a (subinterval of) a side of either $C_v$ or $C_w$. The former polygons have at most three sides each, so the sides of $C$ fall on one of the at most six lines given by these polygon sides. Moreover, $C$ is convex thus any given line can contain at most one side of $C$. Thus $C$ has at most six sides and therefore at most six vertices.

*Case II: All vertices of $C_{\text{opt}}$ are inside $D^*$.*

We will show that we can generate a polynomial-size grid set $\mathcal{G}$ with the desired properties. Recall that we can generate a set $\mathcal{B}$ of candidate circumscribed ellipses of $Q$ in time $O(|X|^5) = O(n^{10})$ time, that is, for all $t$-tuples of $X$ with $t \leqslant 5$ we compute the corresponding circumscribed ellipse and the affine transform to make the ellipse into a circle. We set the circle's radius as unit and fix a coordinate system centered at the circle center. Partition the square $\sigma = [-32/\varepsilon, 32/\varepsilon] \times [-32/\varepsilon, 32/\varepsilon]$ into a regular grid $G$ consisting of $k \times k$ cells, where $k = 2^{16}/\varepsilon^3$; here we assume for simplicity and without loss of generality that $1/\varepsilon$ is integral. Note that the cells have side length $\varepsilon^2/2^{10}$. For each of the $t$-tuples, generating the grid $G$ takes $O(1/\varepsilon^6)$ time. After generating the grid, we transform it back to the original coordinate system.

Let $\mathcal{G}$ be the set of $O(n^{10})$ grids generated this way. Notice that creating all grids $G \in \mathcal{G}$ takes $O(n^{10}/\varepsilon^6)$ time in total. It remains to show that there is an intersecting polygon $C$ with area at most $(1+\varepsilon) \cdot \text{area}(C_{\text{opt}})$ such that $V(C) \subset G$ for some $G \in \mathcal{G}$. To this end, let $G$ be the rectangular grid corresponding to the minimum-area circumscribed ellipse of $Q$. Since this ellipse is defined by at most five points from $Q$, we have $G \in \mathcal{G}$, that is, $G$ is one

of the generated grids. For the remainder of the proof, it will be convenient to go back to the situation in the transformed space, where $G$ is a $k \times k$ square grid (with $k = 2^{16}/\varepsilon^3$) inside the square $\sigma = [-32/\varepsilon, 32/\varepsilon] \times [-32/\varepsilon, 32/\varepsilon]$. Since affine transformations multiply areas by a constant factor and preserve the convexity of polygons, it suffices to argue about the existence of $C$ in this transformed space.

Now consider the set $V(C_{\text{opt}})$ of vertices of the optimal solution $C_{\text{opt}}$ (in the transformed space). For each $v \in C_{\text{opt}}$, take a grid cell containing $v$, and let $C$ be the convex hull of these cells.

Notice that $|V(C)| \leqslant 4|V(C_{\text{opt}})|$. Moreover, if $\delta$ is the cell diameter of $G$, then $C$ is covered by the Minkowski sum $C_{\text{opt}} \oplus B_\delta$, where $B_\delta$ is the disk of radius $\delta$. Thus we have that

$$\begin{aligned}\text{area}(C) &\leqslant \text{area}(C_{\text{opt}} \oplus B_\delta) \\ &= \text{area}(C_{\text{opt}}) + \delta \cdot \text{per}(C_{\text{opt}}) + \text{area}(B_\delta) \\ &\leqslant \text{area}(C_{\text{opt}}) + \delta \cdot 2^8/\varepsilon + \delta^2\pi,\end{aligned}$$

where the last inequality uses that, since $C_{\text{opt}}$ is convex and $C_{\text{opt}} \subset \sigma$, we have $\text{per}(C_{\text{opt}}) \leqslant \text{per}(\sigma) = 2^8/\varepsilon$. Because $\text{area}(C_{\text{opt}}) \geqslant \text{area}(E') = \pi/4$ and $\delta = \sqrt{2} \cdot (\varepsilon^2/2^{10}) < \varepsilon^2/2^9$, we thus have:

$$\text{area}(C) \leqslant \text{area}(C_{\text{opt}}) + 2^8\delta/\varepsilon + \delta^2\pi < \text{area}(C_{\text{opt}}) + \varepsilon/2 + \varepsilon/2^{16} < (1+\varepsilon)\text{area}(C_{\text{opt}}),$$

concluding the proof. □

## 3 An FPTAS for the Minimum-Perimeter Problem of Convex Objects in the Plane

Let $\mathcal{O}$ be a set of convex polygons in the plane of total complexity $n$ for which we want to compute a minimum-perimeter convex intersecting polygon. We assume that $\mathcal{O}$ cannot be stabbed by a single point—this can be verified in $O(n)$ time (i.e., without increasing the running time). To see why, notice that the problem is equivalent to deciding if all of the $n$ half-planes given by the polygon sides (and covering the polygons) have a common intersection, which is a 2-dimensional linear program with $n$ constraints, which can be solved in $O(n)$ time [18]. Since a minimum-perimeter intersecting polygon is necessarily convex, we will from now on drop the adjective "convex" from our terminology. We do this even when referring to convex intersecting polygons that are not necessarily of minimum-perimeter.

In the previous section we have seen that for any $\varepsilon > 0$, we can find a feasible rectangle $R$ and a square $\sigma$ with the following property: $\text{per}(R) \leq (1+\varepsilon)\text{OPT}$, or $C_{\text{opt}} \subseteq \sigma$ with $\text{diam}(C_{\text{opt}}) = \Omega(\text{diam}(\sigma))$. Next we describe an algorithm that, given a parameter $\varepsilon > 0$ and a corresponding square $\sigma$, outputs an intersecting polygon $C^* \subseteq \sigma$ for $\mathcal{O}$ such that if $\text{per}(R) \geq (1+\varepsilon)\text{per}(C_{\text{opt}})$, then $\text{per}(C^*) \leqslant (1+\varepsilon)\text{OPT}$, where $\text{OPT} = \text{per}(C_{\text{opt}})$ (cf. Lemma 4). Finally, we output either $R$ or $C^*$, whichever has smaller perimeter.

Our algorithm starts by partitioning $\sigma$ into a regular grid $G(\sigma)$ of $O(1/\varepsilon^2)$ cells of edge length at most $(\varepsilon/8)\cdot\text{OPT}$. We say that a convex polygon is a *grid polygon* if its vertices are grid points from $G(\sigma)$. The following observation is standard, but for completeness we include a proof.

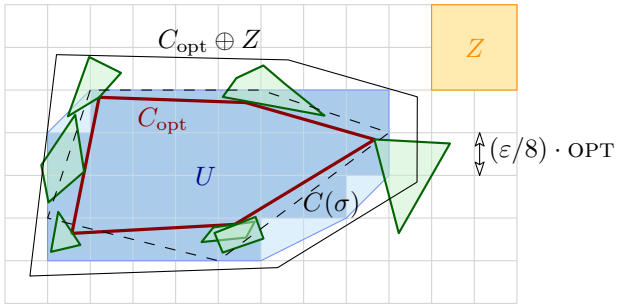



Figure 4: The minimum perimeter intersecting grid polygon $C(\sigma)$ (black, dashed) for a set of input objects (green), the true optimum $C_{\text{opt}}$ (red), and the Minkowski sum $C_{\text{opt}} \oplus Z$ (black, continuous). The darker blue cells represent $U$; their convex hull is sandwiched between $C(\sigma)$ and $C_{\text{opt}} \oplus Z$.

**Observation 9.** *Suppose $\sigma$ contains an optimal solution $C_{\text{opt}}$. Let $C(\sigma)$ be a minimum-perimeter grid polygon that is an intersecting polygon for $\mathcal{O}$. Then* $\text{per}(C(\sigma)) \leqslant (1+\varepsilon) \cdot \text{OPT}$.

*Proof.* Let $U$ be the union of all grid cells that intersect $C_{\text{opt}}$, and let $Z$ be a square of edge length $(\varepsilon/4) \cdot \text{OPT}$, centered at the origin; see Figure 4. Then $U$ is contained in the Minkowski sum $C_{\text{opt}} \oplus Z$. Since $C_{\text{opt}} \oplus Z$ is convex, this implies that the convex hull $\text{CH}(U)$ of $U$, which is a grid polygon that is an intersecting polygon for $\mathcal{O}$, is contained in $C_{\text{opt}} \oplus Z$. Because the perimeter of a Minkowski sum of two convex objects equals the sum of the perimeters of the objects, we have

$$\text{per}(C(\sigma)) \leqslant \text{per}(\text{CH}(U)) \leqslant \text{per}(C_{\text{opt}}) + \text{per}(Z) = (1+\varepsilon) \cdot \text{OPT}. \qquad \square$$

Next we describe an algorithm to compute a minimum-perimeter grid polygon $C(\sigma)$ that is an intersecting polygon for $\mathcal{O}$.

First, we "guess" the lexicographically smallest vertex $v_{\text{bot}}$ of $C(\sigma)$, see Figure 5(i). We can guess $v_{\text{bot}}$ in $O(1/\varepsilon^2)$ different ways. For each possible guess we will find the best solution (if it exists), and then we report the best solution found over all guesses.

Now consider a fixed guess for the lexicographically smallest vertex $v_{\text{bot}}$ of $C(\sigma)$. With a slight abuse of notation we will use $C(\sigma)$ to denote a minimum-perimeter grid polygon that is an intersecting polygon of $\mathcal{O}$ and that has $v_{\text{bot}}$ as lexicographically smallest vertex. (If the polygon $C(\sigma)$ does not exist, the algorithm described below will detect this.) We will compute $C(\sigma)$ via a DP.

The vertices of $C(\sigma)$ are grid points in the region $h^+ \setminus \rho_0$, where $h^+$ is the closed half-plane above the horizontal line through $v_{\text{bot}}$ and $\rho_0$ is the horizontal ray emanating from $v_{\text{bot}}$ and pointing to the left. Let $V$ be the set of such grid points, excluding $v_{\text{bot}}$. We first order the points from $V$ in angular order around $v_{\text{bot}}$. More precisely, for a point $v \in V$, let $\phi(v)$ denote the angle over which we have to rotate $\rho_0$ in clockwise order until we hit $v$. For two points $v, w \in V$, we write $v \prec w$ if $\phi(v) < \phi(w)$. Let $V^+ := V \cup \{v_{\text{bot}}, \overline{v}_{\text{bot}}\}$, where $\overline{v}_{\text{bot}}$ is a copy of $v_{\text{bot}}$, and define $v_{\text{bot}} \prec v$ and $v \prec \overline{v}_{\text{bot}}$ for all $v \in V$. The copy

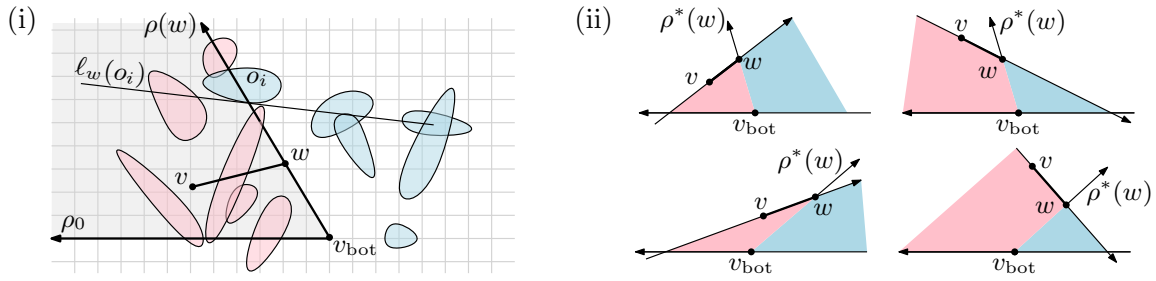



Figure 5: (i) The wedge defined by $\rho_0$ and $\rho(w)$ is shown in light grey. Objects in $\mathcal{O}(v, w)$ are red, objects in $\mathcal{O} \setminus \mathcal{O}(v, w)$ are blue. (ii) The partial solution $C(\Gamma)$ must be contained in the red region, while $C \setminus C(\Gamma)$ must lie in the blue region. There are four situations, depending on whether the angle between $\rho_0$ and $\rho(w)$ is acute or not, and whether the line containing $vw$ intersects $\rho_0$ or not.

$\overline{v}_{\text{bot}}$ serves to distinguish the start and the end vertex of the clockwise circular sequence of vertices of $C(\sigma)$. Note that if $v_{\text{bot}}, v_1, \ldots, v_k, \overline{v}_{\text{bot}}$ denotes this circular sequence, then $v_{\text{bot}} \prec v_1 \prec \cdots \prec v_k \prec \overline{v}_{\text{bot}}$ (since $C(\sigma)$ will never have two vertices that make the same angle with $v_{\text{bot}}$).

We now describe our dynamic-programming algorithm. Consider a polyline from $v_{\text{bot}}$ to some point $v \in V$. We say that this polyline is a *convex chain* if, together with the line segment $vv_{\text{bot}}$, it forms a convex polygon. We denote the convex polygon induced by such a chain $\Gamma$ by $C(\Gamma)$. The problem we now wish to solve is as follows:

> Compute a minimum-length convex chain $\Gamma^*$ from $v_{\text{bot}}$ to $\overline{v}_{\text{bot}}$ such that $C(\Gamma^*)$ is an intersecting polygon for $\mathcal{O}$.

Our dynamic-programming algorithm uses the partial order $\prec$ defined above. We now want to define a subproblem for each point $v \in V^+$, which is to find the "best" chain $\Gamma_v$ ending at $v$. For this to work, we need to know which objects from $\mathcal{O}$ should be covered by the partial solution $C(\Gamma_v)$. This is difficult, however, because objects that intersect the ray from $v_{\text{bot}}$ and going through $v$ could either be intersected by $C(\Gamma_v)$ or by the part of the solution that comes after $v$. To overcome this problem we let the subproblems be defined by the last edge on the chain, instead of by the last vertex. This way we can decide which objects should be covered by a partial solution, as explained next.

Consider a convex chain from $v_{\text{bot}}$ to a point $w \in V^+$ whose last edge is $vw$. Let $\rho(w)$ be the ray emanating from $v_{\text{bot}}$ in the direction of $w$, and let $\rho^*(w)$ be the part of the ray starting at $w$. For $w = \overline{v}_{\text{bot}}$ we define $\rho(w)$ to be the horizontal ray emanating from $v_{\text{bot}}$ and going to the right, and we define $\rho^*(w) = \rho(w)$. For an object $o_i \in \mathcal{O}$ that intersects $\rho^*(w)$, let $\ell_w(o_i)$ be a line that is tangent to $o_i$ at the first intersection point of $\rho^*(w)$ with $o_i$. We now define the set $\mathcal{O}(v, w)$ to be the subset of objects $o_i \in \mathcal{O}$ such that at least one of the following conditions is satisfied; see also Figure 5(i).

(i) $o_i$ intersects the wedge defined by $\rho_0$ and $\rho(w)$, but not $\rho(w)$ itself; or

(ii) $o_i$ intersects $v_{\text{bot}}w$; or

(iii) $o_i$ intersects $\rho^*(w)$ but not $v_{\text{bot}}w$, and the tangent line $\ell_w(o_i)$ intersects the ray containing $vw$ and ending at $w$.

The next lemma shows that we can use the sets $\mathcal{O}(v,w)$ to define our subproblems.

**Lemma 10.** *Let $C$ be any convex polygon that is an intersecting polygon for $\mathcal{O}$ and that has $v_{\text{bot}}$ as lexicographically smallest vertex and $vw$ as one of its edges. Let $\Gamma_w$ be the part of $\partial C$ from $v_{\text{bot}}$ to $w$ in clockwise order. Then all objects in $\mathcal{O}(v,w)$ intersect $C(\Gamma_w)$ and all objects in $\mathcal{O} \setminus \mathcal{O}(v,w)$ intersect $C \setminus C(\Gamma_w)$.*

*Proof.* Because $C$ is convex, it must lie in $h^+$ (the closed half-plane above the horizontal line though $v_{\text{bot}}$) and in the region to the right of the line $\ell(v,w)$ through $v$ and $w$ and directed from $v$ to $w$. This region is split into two regions by $\rho(w)$; the region to the left of $\rho(w)$ contains $C(\Gamma_w)$ and the region to the right of $\rho(w)$ contains $C \setminus C(\Gamma_w)$. See Figure 5(ii) for various possible configurations.

Consider an object $o_i \in \mathcal{O}(v,w)$. If $o_i$ is in $\mathcal{O}(v,w)$ because of condition (i), then it cannot intersect $C \setminus C(\Gamma_w)$ and, hence, it must intersect $C(\Gamma_w)$. If $o_i$ is in $\mathcal{O}(v,w)$ because of condition (ii), then trivially $o_i$ intersects $C(\Gamma_w)$. Finally, if $o_i$ is in $\mathcal{O}(v,w)$ because of condition (iii), then again it cannot intersect $C \setminus C(\Gamma_w)$, and so it must intersect $C(\Gamma_w)$.

On the other hand, suppose that $o_i \notin \mathcal{O}(v,w)$. If $o_i$ does not intersect $\rho(w)$, then this means that $o_i$ does not intersect the wedge defined by $\rho_0$ and $\rho(w)$. Hence, it cannot intersect $C(\Gamma_w)$ and so it must intersect $C \setminus C(\Gamma_w)$. If $o_i$ intersects $\rho(w)$, then the reason it is not in $\mathcal{O}(v,w)$ is that the tangent line $\ell_w(o_i)$ does not intersect the ray containing $vw$ and ending at $w$, which means the tangent separates $o_i$ from $C(\Gamma_w)$. Hence, $o_i$ must intersect $C \setminus C(\Gamma_w)$. □

We can now state the recursion of our DP. To this end we define, for two points $v, w \in V^+$ with $v \prec w$, a table entry $A[v,w]$ as follows.

$$A[v,w] := \text{the minimum length of a convex chain } \Gamma \text{ from } v_{\text{bot}} \text{ to } w \text{ whose last edge is } vw \text{ and such that all objects in } \mathcal{O}(v,w) \text{ intersect } C(\Gamma), \quad (*)$$

where the minimum is $\infty$ if no such chain exists. Lemma 10 implies the following.

**Observation 11.** *Let $\Gamma^*$ be a shortest convex chain from $v_{\text{bot}}$ to $\overline{v}_{\text{bot}}$ such that $C(\Gamma^*)$ is an intersecting polygon for $\mathcal{O}$. Then* $\text{length}(\Gamma^*) = \min\{A[v,\overline{v}_{\text{bot}}] : v \in V \text{ and } \mathcal{O}(v,\overline{v}_{\text{bot}}) = \mathcal{O}\}$.

*Proof.* Consider any $v \in V$ such that $\mathcal{O}(v,\overline{v}_{\text{bot}}) = \mathcal{O}$. By definition of $A[v,\overline{v}_{\text{bot}}]$, there is a convex chain $\Gamma$ from $v_{\text{bot}}$ to $\overline{v}_{\text{bot}}$ such that $C(\Gamma)$ is an intersecting polygon for $\mathcal{O}$. Hence, $\text{length}(\Gamma^*) \leqslant \min\{A[v,\overline{v}_{\text{bot}}] : v \in V \text{ and } \mathcal{O}(v,\overline{v}_{\text{bot}}) = \mathcal{O}\}$.

Conversely, let $\Gamma^*$ be a minimum-length convex chain from $v_{\text{bot}}$ to $\overline{v}_{\text{bot}}$ such that $C(\Gamma^*)$ is an intersecting polygon for $\mathcal{O}$. Let $v$ be the vertex preceding $\overline{v}_{\text{bot}}$ on $\Gamma^*$. By Lemma 10 all objects in $\mathcal{O} \setminus \mathcal{O}(v,\overline{v}_{\text{bot}})$ intersect $C(\Gamma^*) \setminus C(\Gamma^*_{\overline{v}_{\text{bot}}})$, where $\Gamma^*_{\overline{v}_{\text{bot}}}$ is the part of $\Gamma^*$ from $v_{\text{bot}}$ to $\overline{v}_{\text{bot}}$. But trivially $\Gamma^* = \Gamma^*_{\overline{v}_{\text{bot}}}$ and so $C(\Gamma^*) \setminus C(\Gamma^*_{\overline{v}_{\text{bot}}}) = \emptyset$. Since $C(\Gamma^*)$ is

an intersecting polygon for $\mathcal{O}$, this implies that $\mathcal{O} \setminus \mathcal{O}(v, \overline{v}_{\text{bot}}) = \emptyset$. Hence, $\mathcal{O} = \mathcal{O}(v, \overline{v}_{\text{bot}})$, and so $\text{length}(\Gamma^*) \geqslant \min\{A[v, \overline{v}_{\text{bot}}] : v \in V \text{ and } \mathcal{O}(v, \overline{v}_{\text{bot}}) = \mathcal{O}\}$. □

Hence, if we can compute all table entries $A[v, w]$, then we have indeed solved our problem. (The observation only tells us something about the value of an optimal solution, but it is easy to modify the dynamic-programming algorithm to keep track of partial solutions, which will yield a valid solution.)

The entries $A[v, w]$ can be computed using the following lemma. Define $\Delta(v_{\text{bot}}, v, w)$ to be the triangle with vertices $v_{\text{bot}}, v, w$.

**Lemma 12.** *Let $v, w \in V^+$ with $v \prec w$. Let $V(v, w)$ be the set of all points $u \in V \cup \{v_{\text{bot}}\}$ with $u \prec v$ such that $u$ lies below the line $\ell(v, w)$ through $v$ and $w$ and such that all objects in $\mathcal{O}(v, w) \setminus \mathcal{O}(u, v)$ intersect $\Delta(v_{\text{bot}}, v, w)$. Then*

$$A[v, w] = \begin{cases} |v_{\text{bot}} w| & \text{if } v = v_{\text{bot}} \text{ and all objects in } \mathcal{O}(v, w) \text{ intersect } v_{\text{bot}} w \\ \infty & \text{if } v = v_{\text{bot}} \text{ and not all objects in } \mathcal{O}(v, w) \text{ intersect } v_{\text{bot}} w \\ |vw| + \min\limits_{u \in V(v,w)} A[u, v] & \text{otherwise} \end{cases}$$

*Proof.* We will prove by induction on $v$ (with respect to the order $\prec$) that the recursive formula in the lemma is correct—namely, the resulting value corresponds to the definition $(*)$ on the previous page—for any $v \prec w$. The first two cases in the formula are the bases cases of the inductive proof. Their correctness immediately follows from the definition of $A[v, w]$, since for $v = v_{\text{bot}}$ we have $\Gamma = v_{\text{bot}} w$ and so $C(\Gamma) = v_{\text{bot}} w$.

To prove the third case, let $\Gamma^*$ be a minimum-length convex chain from $v_{\text{bot}}$ to $w$ whose last edge is $vw$ and such that all objects in $\mathcal{O}(v, w)$ intersect $C(\Gamma^*)$. Let $A[v, w]$ be the value computed by the recursive formula given in the lemma. We must prove that $A[v, w] = \text{length}(\Gamma^*)$.

Let $u^*$ be the vertex preceding $v$ on $\Gamma^*$. Then $u^* \prec v$ and (because of convexity) $u^*$ must lie below $\ell(v, w)$. Now consider an object $o_i \in \mathcal{O}(v, w) \setminus \mathcal{O}(u^*, v)$. By Lemma 10 it must be intersected by $\Delta(v_{\text{bot}}, v, w)$, since $\Delta(v_{\text{bot}}, v, w) = C(\Gamma^*) \setminus C(\Gamma^* \setminus vw)$. Hence, $u^* \in V(v, w)$. Moreover, $A[u^*, v] = \text{length}(\Gamma^* \setminus vw)$ by induction. Hence,

$$A[v, w] := |vw| + \min_{u \in V(v,w)} A[u, v] \leqslant |vw| + A[u^*, v] = \text{length}(\Gamma^*).$$

On the other hand, for any $u \prec v$ with $A[u, v] \neq \infty$ there is, by induction, a convex chain of length $A[u, v]$ starting at $v_{\text{bot}}$ whose last edge is $uv$ that intersects all objects in $\mathcal{O}(u, v)$. If $u \in V(v, w)$, then we can extend this chain to a convex chain $\Gamma$ starting at $v_{\text{bot}}$ whose last edge is $vw$ and such that all objects in $\mathcal{O}(v, w)$ intersect $C(\Gamma)$. Hence, for any $u \in V(v, w)$ there is a convex chain $\Gamma$ from $v_{\text{bot}}$ to $w$ whose least edge is $vw$ and such that $C(\Gamma)$ intersects all objects in $\mathcal{O}(v, w)$. Thus $A[v, w] := |vw| + \min\limits_{u \in V(v,w)} A[u, v]$ is at least the minimum length of such a chain.

This finishes the correctness proof of the third case. □

Putting everything together, we can finish the proof of Theorem 1.

*Proof of Theorem 1.* We first use algorithm $A1$ from [8] to compute the rectangle $R$ and the square $\sigma$, which as discussed can be done in $O(n^{2.373}/\varepsilon)$ time. For a square $\sigma$ we guess the vertex $v_{\text{bot}}$ in $O(1/\varepsilon^2)$ different ways.

For each guess we run the dynamic-programming algorithm described above. There are $O(1/\varepsilon^4)$ entries $A[u,v]$ in the dynamic-programming table. The most time-consuming computation of a table entry is in the third branch of $A[v,w]$. Here we need to compute the set $\mathcal{O}(v,w)$, which can be done in $O(n)$ time by checking every $o_i \in \mathcal{O}$. For each of the $O(1/\varepsilon^2)$ points with $u \prec v$ such that $u$ lies below the line $\ell(v,w)$ we then check in $O(n)$ time if all objects in $\mathcal{O}(v,w) \setminus \mathcal{O}(u,v)$ intersect $\Delta(v_{\text{bot}},v,w)$, so that we can compute $A[v,w]$. Hence, computing $A[v,w]$ takes $O(n/\varepsilon^2)$ time, which implies that the whole DP needs $O(n/\varepsilon^6)$ time.

Thus the algorithm takes $O(n^{2.373}/\varepsilon) + O(1/\varepsilon^2) \cdot O(n/\varepsilon^6) = O(n^{2.373}/\varepsilon + n/\varepsilon^8)$ time. □

*Remark* 13. We believe that variants of Theorem 1 will also hold when $\mathcal{O}$ is a set of constant-complexity convex objects such as disks. The ingredients that are needed are:

(i) Lemma 4;
(ii) an algorithm A1 that, for a given $\varepsilon_1 > 0$, outputs a rectangle $R$ intersecting all input objects in $\mathcal{O}$ and with perimeter at most $\frac{4}{\pi}(1+\varepsilon_1)\text{OPT}$;
(iii) an algorithm to decide if a point lies inside a given object and a polynomial algorithm to decide whether all objects in $\mathcal{O}$ can be stabbed by a single point.

Note that Lemma 4 has already been proven for arbitrary convex objects by Dumitrescu and Jiang [8]. Moreover, the algorithms in (iii) are typically easy to obtain for constant-complexity objects. (Deciding if there is a point stabbing all objects, for instance, can be done by constructing the arrangement induced by the objects and then checking if there is a face contained in all objects.) Hence, the only thing that is still needed is an algorithm A1 as mentioned in (ii). Dumitrescu and Jiang [8] provided an LP-based algorithm A1 for polygonal objects. We expect that for non-polygonal objects, a polynomial algorithm should be feasible as well.

## 4 An FPTAS for the Minimum-Area Convex Intersecting Polygon

Due to Lemma 7, either there exists an approximate solution with at most six vertices, or it is sufficient to compute the minimum-area convex intersecting polygon whose vertices are in a grid $G \in \mathcal{G}$. Since we have no way to distinguish between these outcomes, we will compute a minimum feasible solution of at most six vertices, as well as a minimum feasible solution (if it exists) in each grid $G \in \mathcal{G}$, and simply return the smallest area polygon that we have found.

In Section 4.1 we will show how to compute an (approximate) minimum-area polygon with at most six vertices with known algebraic methods in $O(n^{13}\log(b/\varepsilon))$ time. Before we do so, we explain how to adapt our dynamic-programming algorithm from Section 3 for computing a minimum-perimeter solution in a grid $G \in \mathcal{G}$ so that it computes a minimum-

area solution. To this end, let us fix a grid $G \in \mathcal{G}$. Keeping the notations as before, we see that everything up to (and including) Lemma 10 still holds. We define the subproblems as follows.

$A[v,w]$ := the minimum area of the convex hull $C(\Gamma)$ for a convex chain $\Gamma$ from $v_{\text{bot}}$ to $w$ whose last edge is $vw$ and such that all objects in $\mathcal{O}(v,w)$ intersect $C(\Gamma)$,

where the minimum is $\infty$ if no such chain exists. The proof of the following observation is the same as the proof of Observation 11, one only needs to change all mentions of length or perimeter of a convex chain to the area of the convex hull of the chain.

**Observation 14.** *Let $\Gamma^*$ be a convex chain from $v_{\text{bot}}$ to $\overline{v}_{\text{bot}}$ where $C(\Gamma^*)$ has minimum area and it is an intersecting polygon for $\mathcal{O}$. Then* $\text{area}(C(\Gamma^*)) = \min\{A[v,\overline{v}_{\text{bot}}] : v \in V$ *and* $\mathcal{O}(v,\overline{v}_{\text{bot}}) = \mathcal{O}\}$.

The recursion also works analogously: we only need to change the starting value, and in the recursive step we add the area of the triangle corresponding to the new segment instead of its length. The proof is again analogous to the minimum-area variant (to Lemma 12).

**Lemma 15.** *Let $v, w \in V^+$ with $v \prec w$. Let $V(v,w)$ be the set of all points $u \in V \cup \{v_{\text{bot}}\}$ with $u \prec v$ such that $u$ lies below the line $\ell(v,w)$ through $v$ and $w$ and such that all objects in $\mathcal{O}(v,w) \setminus \mathcal{O}(u,v)$ intersect $\Delta(v_{\text{bot}},v,w)$. Then*

$$A[v,w] = \begin{cases} 0 & \text{if } v = v_{\text{bot}} \text{ and all objects in } \mathcal{O}(v,w) \text{ intersect } v_{\text{bot}}w \\ \infty & \text{if } v = v_{\text{bot}} \text{ and not all objects in } \mathcal{O}(v,w) \text{ intersect } v_{\text{bot}}w \\ \text{area}(\Delta(v_{\text{bot}},v,w)) + \min\limits_{u \in V(v,w)} A[u,v] & \text{otherwise.} \end{cases}$$

Putting everything together, we can prove Theorem 2.

*Proof.* First, we compute the approximately optimal intersecting polygon with at most six vertices in $O(n^{13}\log(b/\varepsilon))$ time by Theorem 17, see below. The set $\mathcal{G}$ of $O(n^{10})$ candidate grids can be computed in $O(n^{10}/\varepsilon^6)$ time. Each grid $G \in \mathcal{G}$ has $O(1/\varepsilon^6)$ vertices, among which we guess the vertex $v_{\text{bot}}$ in $O(1/\varepsilon^6)$ time.

For each guess we run the dynamic programming algorithm, whose table has $O(1/\varepsilon^{12})$ entries $A[u,v]$. Again the most time-consuming part of computing a table entry is in the third case of Lemma 15: for each of the $O(1/\varepsilon^6)$ points $u \in V(v,w)$ we check if all objects in $\mathcal{O}(v,w) \setminus \mathcal{O}(u,v)$ intersect $\Delta(v_{\text{bot}},v,w)$. Thus computing $A[v,w]$ takes $O(n/\varepsilon^6)$ time, and the DP takes $O(n/\varepsilon^{18})$ time. This part of the algorithm therefore takes $O(n^{10}) \cdot O(1/\varepsilon^6) \cdot O(n/\varepsilon^{18}) = O(n^{11}/\varepsilon^{24})$ time. □

### 4.1 Approximating the minimum-area polygon of constantly many vertices

In this section we will explain how to compute an approximate minimum-area intersecting polygon with $k \leqslant 6$ vertices.. If $k \leqslant 2$, then the optimum is a degenerate polygon (a segment

or a point) of area 0. Since the length of the segment is irrelevant, it is in fact enough to check if there is a line that intersects all input objects. If there is such a line, then it can be rotated until it passes through two vertices of input objects without affecting feasibility. Thus, in $O(n^2)$ time, we can enumerate all pairs of vertices, and for each pair we can check in $O(n)$ time whether the line through them intersects all objects. Consequently, in $O(n^3)$ time we can decide if there is a feasible solution with $k \leqslant 2$ (and $\textsc{opt} = 0$).

Assume now that $\textsc{opt} > 0$ and $3 \leqslant k \leqslant 6$.

We represent the polygon whose vertices listed in clockwise order along the boundary are $v_0, \ldots, v_{k-1}$, by the point $(x_1, y_1, \ldots, x_k, y_k) \in \mathbb{R}^{2k}$; here $x_i$ and $y_i$ denote the $x$- and $y$-coordinate, respectively, of $v_i$. Since we are interested in polygons on *at most* six vertices, it will be convenient to allow consecutive vertices along the boundary to coincide, and consecutive triplets to be collinear. This way we can fix $k = 6$ and work in $\mathbb{R}^{2k} = \mathbb{R}^{12}$, while still being able to represent polygons with less than six vertices.

Let $C(q)$ denote the polygon represented by a point $q = (x_0, y_0, \ldots, x_5, y_5) \in \mathbb{R}^{12}$. Note that not every point $q \in \mathbb{R}^{12}$ corresponds to a convex polygon. We are now looking for a point $q$ such that

(i) $C(q)$ is convex,
(ii) $C(q)$ intersects each $o \in \mathcal{O}$,
(iii) $C(q)$ has minimum area among all points in $\mathbb{R}^{12}$ satisfying (i) and (ii).

First, we change this minimization problem into a decision problem. Thus we are given a parameter $\mu > 0$, and we replace condition (iii) by the condition $\text{area}(C(q)) \leqslant \mu$. We will now show that these conditions are semi-algebraic—they can be expressed as constant-degree polynomial inequalities over the variables $x_i$ and $y_i$—so that the feasible region is a semi-algebraic set. This will allow us to use a result of Basu *et al.* [3] to solve the decision problem, which is to check if the defined semi-algebraic set is empty, in polynomial time.

The convexity of $C(q)$ can be expressed by the fact that consecutive triplets of vertices $v_j, v_{j+1}, v_{j+2}$ (with indices taken modulo 6) have the same orientation. Thus, $C(q)$ is convex if and only if

$$\bigwedge_{j \in [6]} \Big( \det(v_j - v_{j+1},\ v_j - v_{j+2}) \geqslant 0 \Big),$$

where indices are modulo 6 and $\det(a, b)$ is the determinant of the $2 \times 2$ matrix with columns $a$ and $b$. This test is the conjunction of six polynomial inequalities of degree 2.

To check that the polygon defined by $v_0, \ldots, v_5$ intersects a given polygon $o \in \mathcal{O}$, we proceed as follows. Notice that if $C(q)$ intersects $o$, then either $\partial C(q)$ intersects $\partial o$, or one of these polygons contains the other in its interior. This can be expressed as a disjunction of several tests: for each pair (side of $C(q)$, side of $o$) we add a test if they intersect, for one (arbitrary) vertex of $o$ we add a test if it is contained in $C(q)$, and we add a test if $(x_0, y_0)$ is contained in $o$.. Each test is again expressible as a Boolean function of the non-negativity of the signs of $2 \times 2$ determinants involving the vertices of $o$ and $C(q)$.

Finally, for a target value $\mu > 0$, we can express the condition $\text{area}(C(q)) \leqslant \mu$ using he shoelace formula [4], which is a degree-2 polynomial of the variables $(x_i, y_i)$.

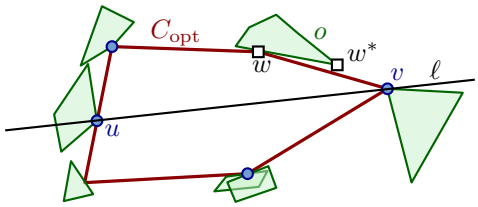



Figure 6: A lower bound for the area of $C_{\text{opt}}$. Blue disk nodes denote the set $X \cap \partial C_{\text{opt}}$. The area of $C_{\text{opt}}$ is greater than the area of the triangle $uvw^*$.

Thus, we can write a Boolean formula $\Phi_\mu$ of $O(n)$ polynomial inequalities of degree 2, where the variables $x_0, y_0, \ldots, x_5, y_5$ are existentially quantified, such that $\Phi_\mu$ is true if and only if there is an intersecting convex polygon $C(q)$ with at most six vertices such that $\text{area}(C(q)) \leqslant \mu$. Basu, Pollack and Roy [3] provide an algorithm to decide the truth of a formula with $t$ existentially quantified variables that has $s$ polynomial inequalities of degree at most $d$ in $s^{t+1}d^{O(t)}$ time, which is $O(n^{13})$ time in case of $\Phi_\mu$. We can therefore use a binary search strategy to find a value $\mu$ such that $\Phi_{\mu/(1+\varepsilon)}$ is false and $\Phi_\mu$ is true. One can observe that it is sufficient to search among the values $\mu = (1+\varepsilon)^z$ where $z$ is an integer. In order to ensure that the strategy can begin and it terminates, it is necessary that we have an upper bound and a strictly positive lower bound for the optimum area OPT.

Recall that the coordinates of the vertices of our input polygons are specified using $b$ bits. By scaling, we get that the points come from a $2^b \times 2^b$ integer grid, and we fix the unit to be the side length of a grid cell. Observe that $\text{OPT} \leqslant (2^b)^2 = 2^{2b}$ as the circumscribed square of the grid is a valid solution. We will now work towards a lower bound.

**Lemma 16.** *Let $C_{\text{opt}}$ be a minimum-area convex intersecting polygon for a set $\mathcal{O}$ of convex polygons. If $\text{area}(C_{\text{opt}}) > 0$, then $\text{area}(C_{\text{opt}}) \geqslant \Omega(\frac{1}{2^{24b}})$.*

*Proof.* If $\text{area}(C_{\text{opt}}) > 0$, then $C_{\text{opt}}$ has at least three edges and thus contains at least two vertices from $X$ by Lemma 6. Let $u, v \in X \cap \partial C_{\text{opt}}$ be such vertices. Let $\ell$ be the line through $u$ and $v$, and let $o$ be an object of $o$ that does not intersect $\ell$, see Figure 6 for an illustration. Note that such an object $o$ exists, as otherwise some segment of $\ell$ would provide a solution of area 0. Let $w$ be a point in $C_{\text{opt}} \cap o$, and let $w^*$ be a vertex of $o$ whose distance to $\ell$ is the minimum among the points of $o$. Clearly $\text{area}(C_{\text{opt}}) \geqslant \text{area}(uvw) \geqslant \text{area}(uvw^*)$. Consequently, it is sufficient to prove the lower bound for the area of $\Delta(uvw^*)$. We claim that the segment $uv$ has length $\Omega(1/2^{4b})$, and the distance of $w^*$ to $\ell$ is at least $\Omega(1/2^{16b})$.

Consider first the point $u$. It is either an input point (and thus it has integer coordinates) or it is the intersection of two polygon edges, each of which are defined as the line through two points form the integer $2^b \times 2^b$ grid. The line through the grid points $g, h$ can be written as $(y-g_y)(h_x-g_x)+(x-g_x)(g_y-h_y) = 0$. Therefore, $u$ is the intersection of the lines $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$, where $a_i, b_i, c_i$ $(i = 1, 2)$ are integers between 0 and $2^{2b}$. Elementary geometry yields that the first coordinate $u_x$ of the intersection can be written in the form $p_u/q_u$, where $p_u$ and $q_u$ are quadratic expressions of $a_i, b_i, c_i$ $(i = 1, 2)$ with coefficients $\pm 1$. We also write the first coordinate of $v$ as $p_v/q_v$. Thus if the first coordinate

of $u$ and $v$ are different, then they differ by at least $1/q_u q_v \geqslant 1/O(2^{4b})$, so applying the same argument to the second coordinates, we get $\operatorname{dist}(u,v) \geqslant \min(|u_x - v_x|, |u_y - v_y|) \geqslant \Omega(1/2^{4b})$. In general, we have shown that the distance of two points of $X$ is at least $\Omega(1/2^{4b})$.

Now consider the distance of $w^*$ to $\ell$. The line $\ell$ through $u$ and $v$ can again be written as $(y-u_y)(v_x-u_x)+(x-u_x)(u_y-v_y)=0$, so it can be written as $\alpha x+\beta y+\gamma=0$ where $\alpha, \beta$ and $\gamma$ are at most quadratic in the coordinates of $u$ and $v$. Recall that the coordinates of $u$ and $v$ can be written as fractions of the coordinates of the input points. Hence, as a function of the integer input, $\alpha$ and $\beta$ are fractions of polynomials of degree at most 8. The distance of the integer point $w^*$ to the line $\ell$ is $\frac{|\alpha w_x^* + \beta w_y^* + \gamma|}{\sqrt{\alpha^2+\beta^2}}$. Since the distance is not 0, the numerator must be at least $\Omega(1/2^{8b})$, and the denominator is the square root of a polynomial fraction where the polynomials have constant coefficients and degree at most 16. Thus the distance of $w^*$ to $\ell$ is at least $\Omega(1/2^{8b})/\sqrt{O(2^{16b})} = \Omega(1/2^{16b})$.

The area of the triangle is therefore $\operatorname{area}(\Delta(uvw^*)) = \frac{1}{2}\operatorname{dist}(u,v) \cdot \operatorname{dist}(w^*,\ell) = \Omega(1/2^{24b})$. The same lower bound holds for $\operatorname{area}(C_{\text{opt}})$, which concludes the proof. □

**Theorem 17.** *Let $\mathcal{O}$ be a set of convex polygons with $n$ vertices in total, whose coordinates are specified using at most $b$ bits, and let $\varepsilon > 0$. Suppose that the minimum-area convex intersecting polygon of $\mathcal{O}$ with at most six vertices has area* OPT. *Then given $\varepsilon$ and $\mathcal{O}$, we can compute in $O(n^{13}\log(b/\varepsilon))$ time a convex intersecting polygon $C$ of $\mathcal{O}$ where* $\operatorname{area}(C) \leqslant (1+\varepsilon)$OPT.

*Proof.* Let $\Phi_\mu$ be the formula described above with target value $\mu$. The Boolean formula for any $\mu > 0$ can be evaluated in time $n^{O(1)}$. By our upper bound and by Lemma 16 the range of possible values for OPT is between $\Omega(1/2^{24b})$ and $2^{2b}$, thus the number of distinct values to consider is $O(\log_{1+\varepsilon} \frac{2^{2b}}{1/2^{24b}}) = O(\frac{b}{\varepsilon})$. Doing binary search over this many values leads to $O(\log(b/\varepsilon))$ evaluations of some formula $\Phi_\mu$, which by Basu, Pollack and Roy [3] runs in $O(n^{13})$ time, resulting in an overall running time of $O(n^{13} \cdot \log(b/\varepsilon))$. □

## 5 An Exact Algorithm for the Minimum-Perimeter Intersecting Polygon of Segments

We describe an exact algorithm to compute a minimum-perimeter intersecting polygon for a set $\mathcal{O}$ of line segments[1] in the plane. Consider an optimal solution $C_{\text{opt}}$, and let $Y$ be the set of all endpoints of the segments in $\mathcal{O}$. The exact algorithm is based on two subroutines for the following two problems. As before, whenever we talk about intersecting polygons, we implicitly require them to be convex. Figure 7 illustrates the polygons computed by these subroutines.

- *Subroutine I*: If $\mathcal{O}$ admits a minimum-perimeter intersecting polygon with none of its vertices being in $Y$, then compute such a polygon, and otherwise report $+\infty$.

[1] Although we describe our algorithm for non-degenerate segments, all our arguments also work if some (or all) of the segments are in fact lines, rays, or even points.

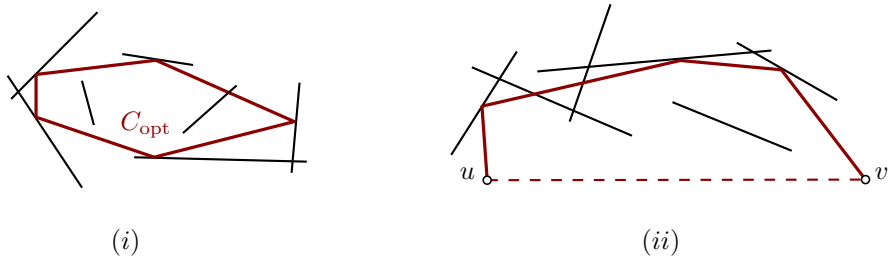



Figure 7: (i) Subroutine I computes a minimum-perimeter intersecting polygon whose vertices are not segment endpoints. (ii) Subroutine II computes a polygon with fixed edge $uv$ for some (sub)set of segments. With the exception of $u$ and $v$, the polygon's vertices are not allowed to be segment endpoints.

- *Subroutine II*: Given two points $u, v \in \mathbb{R}^2$ and a subset of segments $\mathcal{O}' \subseteq \mathcal{O}$, decide if $\mathcal{O}'$ admits an intersecting polygon that has $uv$ as one of its edges and none of whose other vertices belongs to $Y$ and, if so, compute such an intersecting polygon of minimum perimeter. If no such intersecting polygon exists, report $+\infty$. Note that we allow $u = v$, in which case the edge $uv$ degenerates to a point.

### 5.1 Subroutines I and II

The goal of this subsection is to show the following theorem.

**Theorem 18.** *There exist exact algorithms for* Subroutine I *and* Subroutine II *that run in time* $O(n^6 \log n)$ *and* $O(n^3 \log n)$*, respectively.*

We want to establish that computing a portion of a minimum-perimeter polygon that does not contain any segment endpoints is tractable in the sense that we can compute it with a black-box application of [6]. More specifically, we will prove Theorem 18 by invoking an algorithm by Dror, Efrat, Lubiw, and Mitchell [6] for the Touring Polygons Problem as a black box. For completeness we repeat the relevant definitions and results from [6]. We start by defining the Touring Polygons Problem (TPP). Note that in [6] a more general version is considered that requires the solution to remain within a fenced area, but we are only interested in the "unfenced" special case which slightly simplifies notation.

**Definition 19** ((unfenced) Touring Polygons Problem (TPP)[6])**.** In the Touring Polygons Problem (TPP) we are given a sequence of convex polygons $P_1, P_2, \ldots, P_k$, a starting point $s = P_0$, and an ending point $t = P_{k+1}$. We say that a path $\pi$ visits $P_i$, if $\pi \cap P_i \neq \varnothing$, and that $\pi$ visits the polygon sequence $P_1, \ldots, P_k$ if there exist points $q_i \in P_i$ such that the $q_i$'s appear in order of index along $\pi$. We seek to output the shortest path $\pi$ starting at $s$ and visiting the sequence of input polygons before finishing at $t$.

Let $n$ be the total complexity of the $k$ polygons input to TPP (i.e., $n$ is the total number of vertices specifying the $k$ polygons). Then:

**Theorem 20** (Theorem 2 in [6]). *The TPP for arbitrary convex polygons $P_i$ can be solved in time $O(nk^2 \log n)$, using $O(kn)$ space.*

Note that for segments and half-planes, TPP can be solved in $O(n^3 \log n)$ time and quadratic space. An important property of an optimal solution to TPP is *uniqueness*:

**Lemma 21** (Lemma 8 in [6] (unfenced)). *For any points $s, t \in \mathbb{R}^2$ there is a unique shortest path from $s$ to $t$ that visits the polygons $P_1, \dots P_k$ in order.*

**Reduction.** We define a set of half-planes and an ordering on them, such that the optimum tour of the half-planes that respects the ordering gives a minimum intersecting polygon of the segments. In what follows, we will deal with a portion of the boundary of a minimum intersecting polygon, that is, a convex chain $\Gamma$ with fixed endpoints $u$ and $v$. For the sake of simplicity, we rotate the chain so that $u, v$ are on the $x$-axis, and the chain $\Gamma$ lies in the upper half-plane. Let $C(\Gamma)$ denote the (convex) polygon given by $\Gamma$ and the segment $uv$. The set of line segments to be intersected by $C(\Gamma)$ will be denoted by $\mathcal{O}$.

Let us thus fix two points $u, v$ on the $x$-axis, and let $\mathcal{O}$ be a set of $n$ segments. The *half-plane of $o \in \mathcal{O}$ with respect to $u, v$* is the (closed) half-plane defined by $\text{line}(o)$ that (i) does not contain $u$, or (ii) if $u \in \text{line}(o)$, then it is the half-plane containing $v$, or (iii) if $\text{line}(o) = \overline{uv}$, then it is the upper half-plane defined by the $x$-axis. We sort all these half-planes of segments according to the direction of their normal vectors in clockwise order, where the ordering of directions is set to start with $(0, -1)$. Let $\vec{n_i}$ be the normal vector of half-plane $H_i$, and let $\vec{n_0} = (0, -1)$ denote the normal of the half-plane $y \leqslant 0$. Let $o_i$ be the segment that was used to define $H_i$. Finally, let $\mathcal{H}$ denote the set of half-planes $H_i$ for $i \geqslant 1$.

We first show that there exists a convex chain $\Gamma$ from $u$ to $v$ that visits all $H_i$ in order, i.e., there exist $t_i \in H_i$ such that $\Gamma$ visits the $t_i$'s in order. (In fact, we show the stronger statement that this holds for any convex chain from $u$ to $v$ that visits all half-planes.)

**Lemma 22.** *Fix the points $u, v$ on the $x$-axis and let $o_1, \dots, o_n$ be $n$ line segments, with corresponding half-planes $H_i$ where indices follow the ordering of the half-planes according to their normals as above. Assume that there exists a clockwise-oriented convex chain $\Gamma$ from $u$ to $v$ in the half-plane $y \geqslant 0$ that visits each half-plane $H_i$ $(i = 1, \dots, n)$. Then there exist $n$ (not necessarily distinct) points $t_i \in H_i$, such that $\Gamma$ visits the points $t_i$ in the order of their indices.*

*Proof.* For a fixed path $\Gamma$ with the previous properties, the points $t_i$ will be defined with an iterative procedure. Let $j$ be a variable index initially set to one and establish a queue that contains all the half-planes in the specified order. Then the definition of the $t_i$ points comes from the following procedure:

1. Start from the point $u$ and follow the path $\Gamma$ until the first exit from a half-plane of the queue; suppose that this happens at a point $u'$. Let $k$ be the largest index of a half-plane that is being exited at $u'$.
2. Set $t_j, \dots, t_k = u'$.

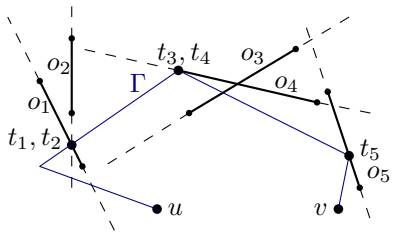



Figure 8: Illustration of the procedure.

3. Set $j$ equal to $k+1$ and remove the half-planes with index less or equal to $k$ from the queue. As long as the queue is not empty, repeat the first step with $u = u'$.

If the point $v$ is part of a half-plane $H_k$, then the path $\Gamma$ would never exit $H_k$. Therefore, there will be an iteration of the previous procedure, such that the path $\Gamma$ in the first step will meet the point $v$, while the queue will not be empty. In this case, the procedure stops and every remaining $t_i$ is set equal to $v$.

Assuming that the $t_i$ points exist, by the construction, the path $\Gamma$ visits them in the specified order. Therefore it is sufficient to show that the points $t_i$ exist and that $t_i \in H_i$ for all $i$. Consider an iteration and let $j$, $k$, $u$ and $u'$ be the corresponding variables of the first two steps of the iteration.

First, we show that for every $i, j \leqslant i \leqslant k$, it holds that $t_i \in H_i$. Suppose to the contrary that $t_i = u' \notin H_i$. Since $\Gamma$ is convex and has clockwise orientation, the rest of $\Gamma$ remains in the interior of the cone defined by the segment $u'v$ and $\partial H_k$ (see Figure 9), which is disjoint from $H_i$ (since (i) $i \leq k$, thus $\sphericalangle(n_0, n_i) \leq \sphericalangle(n_0, n_k)$, and (ii) $u' \notin H_i$). Therefore $\Gamma$ cannot intersect $H_i$ after leaving $u'$. Hence from the feasibility of $\Gamma$, the path must exit $H_i$ before it exits $H_k$, but this contradicts the definition of $u'$.

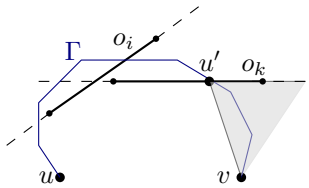



Figure 9: The first exit from a half-plane is not at point $u'$. This is a contradiction.

Next, we show that every half-plane of index larger than $k$ will be visited by $\Gamma$ after the point $u'$. Suppose to the contrary that there exists a half-plane $H_i$ with $i > k$ which is not visited by $\Gamma$ after the point $u'$. In particular, we have that $v \notin H_i$. From the feasibility of $\Gamma$, we can conclude that the path $\Gamma$ has exited from $H_i$ before or exactly at $u'$, but this contradicts the definition of $u'$ and $k$. This concludes the proof. □

If we run the TPP-algorithm from [6] on $u, \mathcal{H}, v$ (where $\mathcal{H}$ is ordered by the order specified above), then we obtain as output the shortest path $P$ from $u$ to $v$ that visits all

half-planes in the given order. It is still possible though for the polygon $C(P)$ defined by $P$ and $uv$ to not intersect $o_i$ (although $P$ visits $H_i$). However we claim that if the optimal solution chain $\Gamma$ for Subroutine I does not go through any segment-endpoint, then $P$ is not only feasible but also optimal for *Subroutine I* on this input. To show this, we need the following lemma, which shows that the convex combination of two paths of different length with the same ordering must be strictly shorter than the longest of the two paths.

**Lemma 23.** *Let* $\Gamma_0 = \{p = a_0, a_1, \ldots, a_n = q\}$, $\Gamma_1 = \{p = b_0, b_1, \ldots, b_n = q\}$ *be two paths such that* $\text{length}(\Gamma_0) < \text{length}(\Gamma_1)$. *For a fixed* $\lambda$, *construct a path* $\Gamma_\lambda = \{p = c_0, c_1, \ldots, c_n = q\}$, *where* $c_i = \lambda \cdot b_i + (1-\lambda) \cdot a_i$. *Then for every* $\lambda, 0 \leqslant \lambda < 1$, *it holds that* $|\Gamma_\lambda| < |\Gamma_1|$.

*Proof.* We can bound $\text{length}(\Gamma_\lambda)$ in the following way:

$$\begin{aligned}
\text{length}(\Gamma_\lambda) &= \sum_{i=0}^{n-1} |c_{i+1} - c_i| \\
&= \sum_{i=0}^{n-1} |\lambda b_{i+1} + (1-\lambda)a_{i+1} - \lambda b_i - (1-\lambda)a_i| \\
&= \sum_{i=0}^{n-1} |\lambda(b_{i+1} - b_i) + (1-\lambda)(a_{i+1} - a_i)| \\
&\leqslant \sum_{i=0}^{n-1} \Big( |\lambda(b_{i+1} - b_i)| + |(1-\lambda)(a_{i+1} - a_i)| \Big) \\
&= \lambda \sum_{i=0}^{n-1} |b_{i+1} - b_i| + (1-\lambda) \sum_{i=1}^{n-1} |a_{i+1} - a_i|. \\
&= \lambda \cdot \text{length}(\Gamma_1) + (1-\lambda)\text{length}(\Gamma_0) \\
&< \text{length}(\Gamma_1).
\end{aligned}$$

□

The next lemma allows us to invoke the algorithm of [6] to compute a polygon portion, assuming that the portion does not contain segment endpoints.

**Lemma 24.** *Let* $C(\Gamma)$ *be a minimum intersecting polygon for a set* $\mathcal{O}$ *of segments. Let* $\mathcal{H}$ *be the set of half-planes corresponding to* $\mathcal{O}$, *and let* $\Gamma_\mathcal{H}$ *be the shortest path from* $u$ *to* $v$ *that visits all half-planes in* $\mathcal{H}$ *in the order defined by their indices, i.e., there are points* $t_i \in H_i$ *so that* $\Gamma_\mathcal{H}$ *goes through the points* $t_i$ *in the order of their indices. Suppose that* $\Gamma$ *does not contain any of the endpoints of the segments (with the possible exception of* $u$ *and* $v$*). Then* $\Gamma = \Gamma_\mathcal{H}$.

*Proof.* By Lemma 22 and the convexity of $\Gamma$, there are $n$ points $a_i \in H_i$ for $\Gamma$ such that the path $\Gamma$ visits them in the specified order. We claim that there are no other vertices on $\Gamma$, that is, it can be represented as the convex chain

$$\Gamma = (u, a_1, \ldots, a_n, v), \text{where } a_i \in H_i.$$

Indeed, if $v$ is a vertex of $\Gamma$ that is not a visit point, then $v$ cannot be the only point where some $o_i$ is visited, so we can shorten $\Gamma$ by shortcutting from the last point of $\bigcup_i o_i$ before $v$ to the first point of $\bigcup_i o_i$ after $v$ (in case $\Gamma$ and some $o_i$ have a shared segment $s$ starting/ending at $v$, then we use the starting/ending point of $s$ as the starting/ending point of the shortcut).

Since $\Gamma$ does not contain endpoints apart from $u$ and $v$, all of its internal vertices are in the interior of some segment in $\mathcal{O}$ that is tangent to $C(\Gamma)$ by a similar shortcutting argument. Notice that if $\Gamma$ visits some segment $o_i$, that is tangent to $C(\Gamma)$, then $H_i$ intersects $\Gamma$ only at this point; consequently, this is the only location where the point $a_i$ could be located. To summarize, at each vertex of $\Gamma$ there must be some segment $o_i$ whose half-plane $H_i$ can be visited only there, and therefore all vertices of $\Gamma$ are contained in the sequence $(u, a_1, \ldots, a_n, v)$.

By the definition of $\Gamma_{\mathcal{H}}$, it contains points $u, b_1, \ldots, b_n, v$ ($b_i \in H_i$) in this order; we claim that it can be represented as the chain

$$\Gamma_{\mathcal{H}} = (u, b_1, \ldots, b_n, v), \text{where } b_i \in H_i.$$

This is easy to see: the chain $(u, b_1, \ldots, b_n, v)$ visits all the half-planes in the order defined by their indices and any tour containing these points in this sequence cannot be shorter than this chain.

Continuing the proof, suppose that $\Gamma \neq \Gamma_{\mathcal{H}}$. The path $\Gamma_{\mathcal{H}}$ is the shortest path visiting the polygons $u, H_1, \ldots, H_n, v$ in this order, so by Lemma 21, $\Gamma_{\mathcal{H}}$ is the unique such shortest path. The path $\Gamma$ is a different chain visiting each of these polygons in the same order, therefore $|\Gamma_{\mathcal{H}}| < |\Gamma|$. Since $\Gamma$ does not visit any endpoint of the segments in $\mathcal{O}$, any convex chain starting at $u$ and ending at $v$ where each point $a_i$ is moved by a small amount towards some other point in $H_i$ is also feasible. In particular, it follows that there exists a small $0 < \lambda < 1$ such that the path

$$\Gamma' = \{u, c_1, \ldots, c_n, v\}, \text{ where } c_i = \lambda a_i + (1 - \lambda) b_i$$

gives a feasible polygon $C(\Gamma')$ for $\mathcal{O}$. By Theorem 23, it holds that $|\Gamma'| < |\Gamma|$, which contradicts the optimality of $\Gamma$. □

We are now ready to prove Theorem 18.

*Proof of Theorem 18.* First we provide an algorithm for *Subroutine I*. If the optimum intersecting polygon's boundary does not go through any segment endpoints, then we use Lemma 24 for an arbitrary vertex $u$ of the optimum as a designated start- and endpoint. It follows that the boundary of the minimum-perimeter intersecting polygon for $\mathcal{O}$ and the optimum tour of the half-planes corresponding to $\mathcal{O}$ in the order defined by their normals is identical.

To find such a point $u$ we use the algorithm from [5]. The algorithm [5] is designed for the so-called *watchman tour* problem, but this is equivalent to our setting, see [15] and the relevant discussion in [6]. The algorithm of [5] computes $O(n^3)$ candidate points (called *event points* in the terminology of [5]) in $O(n^3)$ time, while providing the guarantee that the

optimal tour will go through at least one of them. By Theorem 20 for each of these $O(n^3)$ points a TPP solution can be computed in $O(n^3 \log n)$ time. Then for each TPP solution we can check if the corresponding polygon is convex and intersects all the segments in $O(n^2)$ time. In the end, we return a minimum-perimeter polygon found if it is convex, feasible, and it has no segment endpoints as vertices, and we return $+\infty$ otherwise. That is, we return $+\infty$ if all the minimum-perimeter polygons (among the $O(n^3)$ polygons generated) are non-feasible, non-convex, or contain some segment endpoint. By Lemma 24 any polygon returned by this subroutine is optimal, and its vertices are not segment endpoints. The algorithm takes $O(n^3) \cdot (O(n^3 \log n) + O(n^2)) = O(n^6 \log n)$, time, which concludes the proof for *Subroutine I*.

With respect to *Subroutine II* first note that if $u \neq v$, then all that is required is a single call to the TPP algorithm with a running time of $O(n^3 \log n)$. Furthermore, if $u = v$, then Lemmas 22, 23 and 24 all still work. Consequently, if the optimum tour of the complete set $\mathcal{O}$ of segments contains only one segment endpoint $u$, then the minimum intersecting polygon for the polygons $u, H_1, \ldots H_n, u$ (which is a single call to the TPP algorithm as well) is also optimal for $\mathcal{O}$. Here Lemmas 21, 22, and 24 give a way to detect whether the optimal polygon for a given set of segments and points $u, v$ is equal to the TPP solution. Indeed, if the polygon corresponding to the TPP solution is feasible (the corresponding polygon intersects all segments in $\mathcal{O}$) and its vertices apart from $u, v$ are not from $Y$, then it must be the optimum. Consequently, if the polygon given by the TPP solution is non-convex, or if it fails to intersect some polygon, or if it contains a segment endpoint, then we can return $+\infty$ as the cost of the respective tour. The feasibility checks can be done in $O(n^2)$ time, thus the dominant part of the running time is $O(n^3 \log n)$. □

### 5.2 The proof of Theorem 3

**Setting the stage for the DP.** With Subroutine I available, it remains to find the minimum-perimeter intersecting polygon at least one of whose vertices is a point from $Y$. To this end we will develop an algorithm that, for a given point $p_{\text{bot}} \in Y$, finds a minimum-perimeter intersecting polygon that has $p_{\text{bot}}$ as a vertex (if it exists). We will run this algorithm for each choice of $p_{\text{bot}} \in Y$. Note that when we do so, we may ignore all segments that contain $p_{\text{bot}}$, as they will intersect any intersecting polygon with $p_{\text{bot}}$ as a vertex.

Let $p_{\text{bot}} \in Y$ be given. We first check if $\mathcal{O}$ admits an intersecting polygon that has $p_{\text{bot}}$ as a vertex. This is the case if and only if there exists a half-plane with $p_{\text{bot}}$ on its boundary that overlaps all segments in $\mathcal{O}$. The existence of such a half-plane can easily be tested in $O(n \log n)$ using a rotational sweep around $p_{\text{bot}}$. So from now on we assume that an intersecting polygon exists that has $p_{\text{bot}}$ as a vertex. Let $C_{\text{opt}}$ be a minimum-perimeter such intersecting polygon. The boundary of $C_{\text{opt}}$ consists of chains that connect points from $Y$ and such that the interior vertices of these chains are disjoint from $Y$. (If $p_{\text{bot}}$ is the only point from $Y$ that is a vertex of $C_{\text{opt}}$, then there is a single chain connecting $p_{\text{bot}}$ to itself.) We will develop a dynamic-programming algorithm similar to the algorithm from Section 3. The DP will be based on the points in $Y$ (instead of on grid points) and we will use Subroutine II to find the chains connecting the points from $Y$ on $\partial C_{\text{opt}}$. As we will see, however, there are several challenges that we need to overcome to adapt the DP.

Observe that if we take three rays emanating from $p_{\text{bot}}$ that are at a 120-degree angle from each other, then at least one of them lies fully outside $C_{\text{opt}}$ (except for the starting point $p_{\text{bot}}$). This follows by our assumption that $p_{\text{bot}}$ is a vertex of $C_{\text{opt}}$. Consider such a ray $\rho_0$, and assume without loss of generality that $\rho_0$ is a horizontal ray pointing leftwards. Our goal is now to find a minimum-perimeter intersecting polygon with $p_{\text{bot}}$ as a vertex and that is not intersected by $\rho_0$. As mentioned, we will do this with a dynamic programming algorithm similar to the one from Section 3. The point $p_{\text{bot}}$ will play the role of $v_{\text{bot}}$—but note that $p_{\text{bot}}$ need not be the lexicographically smallest vertex of $C_{\text{opt}}$—and $Y \setminus \{p_{\text{bot}}\}$ will play the role of $V$. Hence, it is convenient to redefine $Y$ as $Y \setminus \{p_{\text{bot}}\}$ and $Y^+ := Y \cup \{p_{\text{bot}}, \overline{p}_{\text{bot}}\}$, where $\overline{p}_{\text{bot}}$ is a copy of $p_{\text{bot}}$. We define a (partial) angular order $\prec$ on $Y^+$, as before.

In Section 3, we knew that $v_{\text{bot}}$ was the lexicographically smallest vertex of $C(\sigma)$, and so we were looking for a solution that lies above the horizontal line through $v_{\text{bot}}$. This was important to be able to decide which objects should be intersected by a partial solution; see Figure 5. In the current setting, however, the point $p_{\text{bot}}$ need not be the lexicographically smallest point. Hence, we also need to guess the orientation of the line tangent to $C_{\text{opt}}$ at $p_{\text{bot}}$. To this end, let $\psi(o_i)$ be the angle over which we have to rotate $\rho_0$ in clockwise order to make it parallel to a given segment $o_i \in \mathcal{O}$, and sort the set of (distinct) angles $\psi(o_i)$ in increasing order. Let $\Psi := \{\psi_1, \psi_2, \ldots\}$, where $\psi_j < \psi_{j+1}$ for all $1 \leqslant j < |\Psi|$, denote this sorted set. Define $\psi_0 := 0$ and $\psi_{|\Psi|+1} = \pi$. The problem we now wish to solve is

> Given a point $p_{\text{bot}}$ and value $j$ with $0 \leqslant j \leqslant |\Psi|$, find a minimum-perimeter intersecting polygon $C_{\text{opt}}$ for $\mathcal{O}$ such that
>
> - $p_{\text{bot}}$ is a vertex of $C_{\text{opt}}$,
> - the horizontal ray $\rho_0$ going from $p_{\text{bot}}$ to the left does not intersect $C_{\text{opt}}$,
> - $C_{\text{opt}}$ has a tangent line $\ell$ at $p_{\text{bot}}$ such that the clockwise angle from $\rho_0$ to $\ell$ lies in the range $[\psi_j, \psi_{j+1}]$.

**The dynamic programming algorithm.** We will now develop our DP for the problem that we just stated, for a given point $p_{\text{bot}} \in Y$, a ray $\rho_0$, and range $[\psi_j, \psi_{j+1}]$; see Figure 10(i).

Let $\mathcal{O}(\rho_0) \subseteq \mathcal{O}$ denote the set of segments that intersect the ray $\rho_0$, and let $\ell_0$ be the line containing $\rho_0$. The line $\ell_0$ may split the optimal solution $C_{\text{opt}}$ into two parts: a part $C^+_{\text{opt}}$ above $\ell_0$ and a part $C^-_{\text{opt}}$ below $\ell_0$. Since we have fixed the range of the tangent at $p_{\text{bot}}$ to lie in the range $[\psi_j, \psi_{j+1}]$, we can split $\mathcal{O}(\rho_0)$ into two subsets, $\mathcal{O}(\rho_0)^+ := \{o_i \in \mathcal{O}(\rho_0) : \psi(o_i) \geqslant \psi_{j+1}\}$ and $\mathcal{O}(\rho_0)^- := \{o_i \in \mathcal{O}(\rho_0) : \psi(o_i) \leqslant \psi_j\}$.

Note that $\mathcal{O}(\rho_0) = \mathcal{O}(\rho_0)^+ \cup \mathcal{O}(\rho_0)^-$, because $\psi_j$ and $\psi_{j+1}$ are consecutive angles in $\Psi$. Because the orientation of the tangent at $p_{\text{bot}}$ lies in the range $[\psi_j, \psi_{j+1}]$, we know that the segments in $\mathcal{O}(\rho_0)^+$ must be intersected by $C^+_{\text{opt}}$, while the segments in $\mathcal{O}(\rho_0)^-$ must be intersected by $C^-_{\text{opt}}$; see Figure 10(i). Intuitively, the segments in $\mathcal{O}(\rho_0)^+$ must be intersected by "the initial part" of $C_{\text{opt}}$, while the segments in $\mathcal{O}(\rho_0)^-$ are intersected by "the later part". We will use this when we define the subproblems in our DP.

In Section 3 we defined subproblems for pairs of grid points $v, w$. The goal of such a subproblem was to find the minimum-length convex chain $\Gamma$ such that $C(\Gamma)$ intersects a

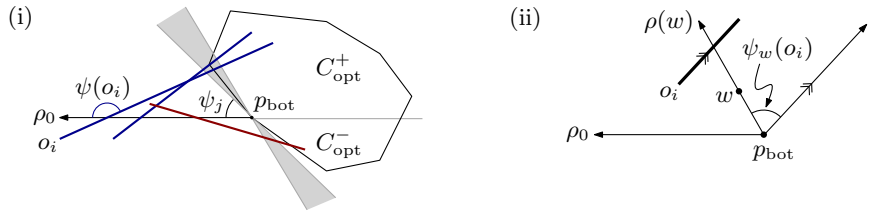



Figure 10: (i) The grey double wedge indicates the region containing the tangent at $p_{\text{bot}}$. (Note that then there must be more segments than the blue and red segments that are shown. In particular, there must be segments parallel to the lines delimiting the double wedge.) The blue segments are in $\mathcal{O}(\rho_0)^+$ so they must be intersected by $C^+_{\text{opt}}$, while the red segment is in $\mathcal{O}(\rho_0)^-$ and must be intersected by $C^-_{\text{opt}}$. (ii) The definition of $\psi_w(o_i)$.

certain subset $\mathcal{O}(v, w)$ and whose last edge is $vw$. The fact that we knew the last edge $vw$ was crucial to define the set $\mathcal{O}(v, w)$, since the slope of $vw$ determined which objects should be intersected by $C(\Gamma)$. In the current setting this does not work: we could define a subproblem for pairs $v, w \in Y$, but "consecutive" vertices $v, w$ from $Y$ along $\Gamma$ are now connected by a polyline $Z_{vw}$ whose inner vertices are disjoint from $Y$. The difficulty is that the polyline $Z_{vw}$ depends on the segments that need to be intersected by $C(\Gamma)$. Hence, there is a cyclic dependency between the set of segments to be intersected by $C(\Gamma)$ (which depends on the slope of $zw$, where $z$ is the vertex of $Z_{vw}$ preceding $w$) and the vertex $z$ (which depends on the segments that need to be intersected by $C(\Gamma)$). We overcome this problem as follows.

Similarly to the previous section, we call a polyline $\Gamma$ from $p_{\text{bot}}$ to some point $v \in Y$ a *convex chain* if, together with the line segment $vp_{\text{bot}}$, it forms a convex polygon. We denote this polygon by $C(\Gamma)$. Consider a convex chain $\Gamma_w$ ending in a point $w \in Y$. Let $\rho(w)$ denote the ray from $p_{\text{bot}}$ through $w$ and let $\rho^*(w)$ be the part of this ray starting at $w$. Let $\mathcal{O}(w)$ be the set of input segments that intersect $\rho^*(w)$. Of those segments, $C(\Gamma_w)$ must intersect the ones such that the line $\ell(o_i)$ containing[2] the segment $o_i$ intersects the ray containing $zw$ and ending at $w$, where $z$ is the (unknown) vertex preceding $w$. For a segment $o_i \in \mathcal{O}(w)$, let $\psi_w(o_i)$ be the angle over which we have to rotate $\rho(w)$ in clockwise order to make it parallel to $o_i$; see Figure 10(ii). Let $\Psi(w) = \{\psi_1, \psi_2, \ldots \psi_j \ldots\}$, for all $1 \leqslant j < |\Psi(w)|$, be the sorted set of (distinct) angles $\psi_w(o_i)$ defined by the segments in $\mathcal{O}(w)$. For an index $j$ with $1 \leqslant j \leqslant |\Psi(w)|$, define $\mathcal{O}(w, j) := \{o_i \in \mathcal{O}(w) : \psi_w(o_i) \leqslant \psi_j\}$, and define $\mathcal{O}(w, 0) = \varnothing$. We call $\mathcal{O}(w, j)$ a *prefix set*. The key observation is that $C(\Gamma_w)$ must intersect the segments from some prefix set $\mathcal{O}(w, j)$, where $j$ depends on the unknown vertex $z$ preceding $w$. So our DP will try all possible prefix sets $\mathcal{O}(w, j)$, and make sure that subproblems are combined in a consistent manner.

We now have everything in place to describe our dynamic-programming table. It consists of entries $A[w, j]$, where $w$ ranges over all points in $Y$, and $j$ ranges over all values for which $\mathcal{O}(w, j)$ is defined. For convenience add two special entries, $A[p_{\text{bot}}, 0]$ and $A[\overline{p}_{\text{bot}}, 0]$; the former will serve as the base case, and the latter will contain (the value of) the final

[2]Since the input objects are now segments, the tangent line $\ell_w(o_i)$ is just the line containing $o_i$.

solution. Note that these are the only ones for $p_{\text{bot}}$ and $\overline{p}_{\text{bot}}$, and that we have at most $|Y| \cdot n = O(n^2)$ entries.

For a pair of distinct rays $\rho_1, \rho_2$ that have a shared starting point, let $W(\rho_1, \rho_2)$ denote the wedge swept by rotating $\rho_1$ anti-clockwise until it coincides with $\rho_2$. Note that $W(\rho_1, \rho_2)$ need not be convex. We will now define the set $\mathcal{O}^*(w, j)$ of segments to be covered in a subproblem.

> For a point $w \in Y$, the set $\mathcal{O}^*(w, j)$ consists of the segments $o_i \in \mathcal{O}$ that satisfy at least one of the following conditions:
>
> (i) $o_i$ intersects the wedge $W(\rho_0, \rho(w))$ but not $\rho(w)$ itself, and $o_i \notin \mathcal{O}^-(\rho_0)$; or
> (ii) $o_i$ intersects $p_{\text{bot}}w$; or
> (iii) $o_i \in \mathcal{O}(w, j)$.
>
> Furthermore, $\mathcal{O}^*(p_{\text{bot}}, 0) := \varnothing$ and $\mathcal{O}^*(\overline{p}_{\text{bot}}, 0) := \mathcal{O}$.

We would like now to define $A[w, j]$ to be the minimum length of a convex chain $\Gamma$ from $p_{\text{bot}}$ to $w$ such that all objects in $\mathcal{O}^*(w, j)$ intersect $C(\Gamma)$. There is, however, a technicality to address: the minimum-perimeter polygon that intersects all segments from $\mathcal{O}$ need not be convex when we require it to have $p_{\text{bot}}$ as a vertex. Indeed, when patching together convex arcs between vertices in $Y$, vertices in $Y$ may be reflex in the resulting polygon. Such a non-convex polygon cannot be a global optimum (as its convex hull is an intersecting polygon of strictly smaller perimeter), implying that $p_{\text{bot}}$ was not the correct choice. This makes a clean definition of our subproblems awkward. Therefore, instead of first defining the subproblems and then giving the recursive formula, we will immediately give the recursive formula and then prove that it computes what we want.

For two points $v, w \in Y^+$ (where $Y^+ = Y \cup \{\overline{p}_{\text{bot}}\}$) with $v \prec w$ and a set $\mathcal{O}' \subseteq \mathcal{O}$, let $L(v, w, \mathcal{O}')$ be the minimum length of a convex chain $\Gamma$ from $v$ to $w$ such that the convex polygon defined by $\Gamma$ and $vw$ is an intersecting set for $\mathcal{O}'$ and all inner vertices of $\Gamma$ are disjoint from $Y$; if no such chain exists then $L(v, w, \mathcal{O}') = \infty$. Recall that we can compute $L(v, w, \mathcal{O}')$ using Subroutine II. As before, let $\Delta(p_{\text{bot}}, v, w)$ denote the triangle with vertices $p_{\text{bot}}, v, w$.

**Definition 25.** Let $w \in Y^+$ and $j$ be a value for which $\mathcal{O}[w, j]$ is defined. Thus $0 \leqslant j \leqslant |\Psi(w)|$, where we set $|\Psi(w)| := 0$ for $w \in \{p_{\text{bot}}, \overline{p}_{\text{bot}}\}$. For $v \prec w$ and $0 \leqslant j' \leqslant |\Psi(v)|$, let

$$\mathcal{O}^*(w, j, v, j') := \mathcal{O}^*(w, j) \setminus \Big(\mathcal{O}^*(v, j') \cup \{o_i \in \mathcal{O} : o_i \text{ intersects } \Delta(p_{\text{bot}}, v, w)\}\Big)$$

and define

$$A[w, j] := \begin{cases} 0 & \text{if } w = p_{\text{bot}} \\ \min\limits_{\substack{v \prec w \\ 0 \leqslant j' \leqslant |\Psi(v)|}} L(v, w, \mathcal{O}^*(w, j, v, j')) + A[v, j'] & \text{otherwise.} \end{cases}$$

The next lemma implies that the table entry $A[\overline{p}_{\text{bot}}, 0]$ defined by this recursive formula gives us what we want. Part (a) implies that $A[\overline{p}_{\text{bot}}, 0]$ will never return a value

that is too small, while part (b) implies that for the correct choice of $p_{\text{bot}}$ and range of orientations for the tangent to $C_{\text{opt}}$ at $p_{\text{bot}}$, the entry $A[\overline{p}_{\text{bot}}, 0]$ gives us (the value of) the optimal solution.

**Lemma 26.** *Consider the table entry $A[\overline{p}_{\text{bot}}, 0]$ defined by Definition 25 for a given point $p_{\text{bot}}$ and range $[\psi_i, \psi_{i+1}]$.*

*(a) There exists a convex intersecting polygon for $\mathcal{O}$ of perimeter at most $A[\overline{p}_{\text{bot}}, 0]$.*

*(b) If $p_{\text{bot}}$ is a vertex of the minimum-perimeter convex intersecting polygon $C_{\text{opt}}$ for $\mathcal{O}$, and $\rho_0$ does not intersect $C_{\text{opt}}$, and there is a tangent line $\ell$ at $p_{\text{bot}}$ whose orientation is in the range $[\psi_i, \psi_{i+1}]$, then $\text{per}(C_{\text{opt}}) = A[\overline{p}_{\text{bot}}, 0]$.*

*Proof.*

(a) Note that the statement trivially holds if $A[\overline{p}_{\text{bot}}, 0] = \infty$. We now prove by induction on $w$ (with respect to the order $\prec$) that for any pair $(w, j)$ for which $A[w, j] < \infty$, there is a sequence $\Xi(w, j) = v_0, v_1, \ldots, v_k$ of points from $Y$, where $p_{\text{bot}} = v_0 \prec v_1 \prec \cdots \prec v_k = w$, with the following property: there are convex polygons $P_1, \ldots, P_k$ such that

- $v_{i-1}v_i$ is an edge of $P_i$,
- $\sum_{i=1}^{k} (\text{per}(P_i) - |v_{i-1}v_i|) = A[w, j]$
- $C(w, j) := \bigcup_{i=1}^{k} (P_i \cup \Delta(p_{\text{bot}}, v_{i-1}, v_i))$ is an intersecting set for $\mathcal{O}^*(w, j)$.

Observe that the perimeter of $C(w, j)$ is at most $A[w, j] + |w\overline{p}_{\text{bot}}|$. Indeed, each edge $v_{i-1}v_i$ is shared between the polygon $P_i$ and the triangle $\Delta(p_{\text{bot}}, v_{i-1}, v_i)$, and so the contribution of the polygons $P_i$ to the perimeter is at most $\sum_{i=1}^{k} (\text{per}(P_i) - |v_{i-1}v_i|) = A[w, j]$. (At most, because the $P_i$'s are not required to be disjoint.) Since the edges $p_{\text{bot}}v_i$ are shared between consecutive triangles $\Delta(p_{\text{bot}}, v_{i-1}, v_i)$ and $\Delta(p_{\text{bot}}, v_i, v_{i+1})$, the contribution of triangles $\Delta(p_{\text{bot}}, v_{i-1}, v_i)$ to the perimeter is at most $|p_{\text{bot}}v_0| + |p_{\text{bot}}v_k| = |p_{\text{bot}}w|$. Thus the perimeter of $C(w, j)$ is at most $A[w, j] + |w\overline{p}_{\text{bot}}|$, as claimed. It follows that the existence of the sequence $\Xi(\overline{p}_{\text{bot}}, 0)$ implies part (a) of the lemma, because $\mathcal{O}^*(\overline{p}_{\text{bot}}, 0) = \mathcal{O}$ and so the convex hull of $C(\overline{p}_{\text{bot}}, 0)$ is a convex intersecting set for $\mathcal{O}$ of perimeter at most $A[\overline{p}_{\text{bot}}, 0]$.

The base case is $(w, j) = (p_{\text{bot}}, 0)$. Since $\mathcal{O}^*(p_{\text{bot}}, 0) = \varnothing$, the sequence consisting of the single point $p_{\text{bot}}$ trivially has the desired properties in this case, with $C(p_{\text{bot}}, 0) = \{p_{\text{bot}}\}$.

Now consider a pair $(w, j)$ with $p_{\text{bot}} \prec w$. Let $j'$ be such that

$$A[w, j] = L(v, w, \mathcal{O}^*(w, j, v, j')) + A[v, j'].$$

By induction, there is a sequence $\Xi(v, j')$ with the properties stated above. We claim that the sequence $\Xi(w, j)$ we obtain by appending $w$ to $\Xi(v, j')$ has the desired properties. Let $\Xi(w, j) = v_0, v_1, \ldots, v_k$ and note that $v_{k-1} = v$ and $v_k = w$. Take $P_k := C(\Gamma)$, where $\Gamma$ is a minimum-length convex chain $\Gamma$ from $v$ to $w$ such that $C(\Gamma)$ is an intersecting set for $\mathcal{O}^*(w, j, v, j')$. The induction hypothesis and the fact

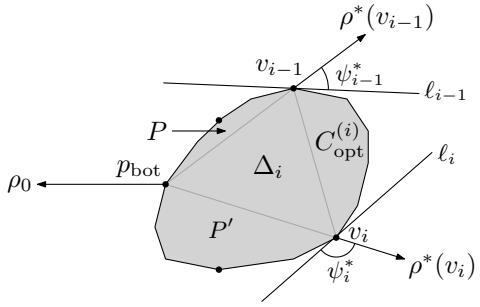



Figure 11: Illustration for the proof of Lemma 26.

that $A[w,j] = L(v,w,\mathcal{O}^*(w,j,v,j')) + A[v,j']$ now implies that $\Xi(w,j)$ has the first two properties. The third property follows from the induction hypothesis and the definition of $\mathcal{O}^*(w,j,v,j')$.

(b) Since $A[\overline{p}_{\text{bot}},0] \geqslant \text{per}(C_{\text{opt}})$ by part (a), it suffices to prove that $A[\overline{p}_{\text{bot}},0] \leqslant \text{per}(C_{\text{opt}})$. (Recall from the statement of the lemma that $C_{\text{opt}}$ refers to the globally optimal solution. This will also be the case throughout the rest of the proof.) To this end, let $p_{\text{bot}} = v_0, v_1, \ldots, v_k = \overline{p}_{\text{bot}}$ be the points from $Y$ on $\partial C_{\text{opt}}$, in clockwise order. Let $\Gamma_i$ denote the part of $\partial C_{\text{opt}}$ from $v_{i-1}$ to $v_i$ (in clockwise order), and let $\Gamma_{\leqslant i}$ be the part of $\partial C_{\text{opt}}$ from $p_{\text{bot}}$ to $v_i$. Let $C^{(i)}_{\text{opt}}$ denote the part of $C_{\text{opt}}$ bounded by the segment $v_{i-1}v_i$ and $\Gamma_i$; see Figure 11 for an illustration.

For $1 \leqslant i < k$, let $\ell_i$ be a line tangent to $C_{\text{opt}}$ at $v_i$, and assume without loss of generality that $\ell_i$ is not parallel to any segment in $\mathcal{O}$. Recall that $\Psi(v_i)$ denotes the sorted sequence of angles $\psi_{v_i}(o)$ defined by the segments in $\mathcal{O}(v_i)$, that is, by the segments intersecting the ray $\rho^*(v_i)$. Let $\psi^*_i$ be the angle over which we have to rotate $\rho(v_i)$ to make it parallel to $\ell_i$. Finally, let $\Psi(v_i) := \langle \psi^{(i)}_1, \psi^{(i)}_2, \ldots \rangle$ and let $j_i$ be the largest index such that $\psi^{(i)}_{j_i} < \psi^*_i$, where we define $j_0 = j_k = 0$.

*Claim.* $C^{(i)}_{\text{opt}}$ intersects all segments from $\mathcal{O}^*(v_i, j_i, v_{i-1}, j_{i-1})$.

*Proof.* Partition $C_{\text{opt}}$ into four pieces: $C^{(i)}_{\text{opt}}$, the triangle $\Delta_i := \Delta(p_{\text{bot}}, v_{i-1}, v_i)$, the piece $P$ given by the convex hull of the clockwise chain from $p_{\text{bot}}$ to $v_{i-1}$, and the piece $P'$ given by the convex hull of the clockwise chain from $v_i$ to $p_{\text{bot}}$. See Figure 11 for an illustration of this partition. Any segment $o \in \mathcal{O}$ intersects at least one of these pieces. We will show that if $o$ intersects $\Delta_i$ or $P$ or $P'$, then $o \notin \mathcal{O}^*(v_i, j_i, v_{i-1}, j_{i-1})$, which implies the claim.

Note that for $i = 1$ we have $v_{i-1} = p_{\text{bot}}$, in which case $\Delta_i$ degenerates to a line segment and $P = \varnothing$. Similarly, for $i = k$ we have $v_i = \overline{p}_{\text{bot}}$, in which case $\Delta_i$ degenerates to a line segment and $P' = \varnothing$. Finally, when $i = k = 1$ we have $C^{(i)}_{\text{opt}} = C_{\text{opt}}$ and the claim trivially holds.

- If $o$ intersects $\Delta_i$ then $o \notin \mathcal{O}^*(v_i, j_i, v_{i-1}, j_{i-1})$ by definition of $\mathcal{O}^*(v_i, j_i, v_{i-1}, j_{i-1})$.

- Now suppose $o$ intersects $P$ but not $\Delta_i$.
  If $o$ intersects $\rho(v_{i-1})$, then it must intersect $\rho^*(v_{i-1})$, since we assumed $o$ does not intersect $\Delta_i$. But if $o$ intersects $P$ and $\rho^*(v_{i-1})$, then $\psi_{v_{i-1}}(o) < \psi^*_{i-1}$ and so $\psi_{v_{i-1}}(o) \leqslant \psi^{(i-1)}_{j_{i-1}}$. This means that $o \in \mathcal{O}(v_{i-1}, j_{i-1}) \subseteq \mathcal{O}^*(v_{i-1}, j_{i-1})$. Hence, $o \notin \mathcal{O}^*(v_i, j_i, v_{i-1}, j_{i-1})$.
  Now assume $o$ does not intersect $\rho(v_{i-1})$. Note that $o$ intersects the wedge from $\rho_0$ to $\rho(v_{i-1})$, since $o$ intersects $P$. Then the only reason for $o$ to not be in $o \in \mathcal{O}^*(v_{i-1}, j_{i-1})$ is if $o \in \mathcal{O}^-(\rho_0)$. But then $o$ cannot intersect $P$. Hence, $o \in \mathcal{O}^*(v_{i-1}, j_{i-1})$ and so $o \notin \mathcal{O}^*(v_i, j_i, v_{i-1}, j_{i-1})$.
- Finally, suppose $o$ intersects $P'$ but not $\Delta_i$. Note that then $v_i \neq \overline{p}_{\text{bot}}$, otherwise $P' = \varnothing$.
  If $o$ intersects $\rho(v_i)$, then it must intersect $\rho^*(v_i)$, since we assumed $o$ does not intersect $\Delta_i$. But if $o$ intersects $P'$ and $\rho^*(v_i)$, then $\psi_{v_i}(o) > \psi^*_i$ and so $\psi_{v_i}(o) > \psi^{(i)}_{j_i}$. This means that $o \notin \mathcal{O}(v_i, j_i)$. Since $o$ intersects $\rho^*(v_i)$, this means that $o \notin \mathcal{O}^*(v_i, j_i)$ and hence, that $o \notin \mathcal{O}^*(v_i, j_i, v_{i-1}, j_{i-1})$.
  If $o$ does not intersect $\rho(v_i)$, then either $o$ lies entirely in the wedge from $\rho(v_i)$ to $\rho_0$ or it intersects $\rho_0$. In the former case $o$ clearly cannot satisfy any of the conditions to be in $\mathcal{O}^*(v_i, j_i)$. In the latter case the fact that $o$ intersects $P'$ implies that $o \in \mathcal{O}^-(\rho_0)$, and so $o \notin \mathcal{O}^*(v_i, j_i)$. Hence, $o \notin \mathcal{O}^*(v_i, j_i, v_{i-1}, j_{i-1})$.

◁

We now prove by induction on $i$ that $A[v_i, j_i] \leqslant \text{length}(\Gamma_{\leqslant i})$ for all $0 \leqslant i \leqslant k$. Since $(v_k, j_k) = (\overline{p}_{\text{bot}}, 0)$ and $\text{length}(\Gamma_{\leqslant k}) = \text{per}(C_{\text{opt}})$, this will prove part (b).

The base case for the induction is when $i = 0$. Then we have $A[v_0, j_0] = A[p_{\text{bot}}, 0] = 0 = \text{length}(\Gamma_0) = \text{length}(\Gamma_{\leqslant 0})$, since $\Gamma_0 = \Gamma_{\leqslant 0} = \{p_{\text{bot}}\}$. Now suppose $i > 0$. Then

$$\begin{aligned}
A[v_i, j_i] &\leqslant L(v_{i-1}, v_i, \mathcal{O}^*(v_i, j_i, v_{i-1}, j_{i-1})) + A[v_{i-1}, j_{i-1}] && \text{(by definition of } A[v_i, j_i]) \\
&\leqslant \text{length}(\Gamma_i) + A[v_{i-1}, j_{i-1}] && \text{(by the claim above)} \\
&\leqslant \text{length}(\Gamma_i) + \text{length}(\Gamma_{\leqslant i-1}) && \text{(by the induction hypothesis)} \\
&= \text{length}(\Gamma_{\leqslant i}),
\end{aligned}$$

which finishes the proof. □

**Putting everything together.** Lemma 26 implies that after solving the DPs for all choices of $p_{\text{bot}}$ and the range $[\psi_i, \psi_{i+1}]$, we have found the minimum-perimeter intersecting set for $\mathcal{O}$. (Computing the intersecting set itself, using the relevant dynamic-programming table, is then routine.)

We can analyse the running time as follows. The number of DPs solved is $O(|Y| \cdot n) = O(n^2)$. The dynamic-programming tables have $O(n^2)$ entries. Computing an entry takes $O(|Y| \cdot n) = O(n^2)$ calls to Subroutine II, at $O(n^3 \log n)$ time each. The DPs thus take $O(n^2) \cdot O(n^2) \cdot O(n^2) \cdot O(n^3 \log n) = O(n^9 \log n)$ time. If the optimal solution does not go through any point of $Y$, then by Theorem 18 it will be found in $O(n^6 \log n)$ time. The optimum of these two algorithms is the global optimum.

This concludes the proof of Theorem 3. □

## 6 Conclusion

We gave fully polynomial time approximation schemes for the minimum-perimeter and minimum-area convex intersecting polygon problems for convex polygons. Additionally, we developed a polynomial-time algorithm for the minimum-perimeter problem of segments.

It is likely that the running times of our algorithms can be improved further. One could also try to generalize the set of objects, for example, adapting the minimum area algorithm to arbitrary convex objects. We propose the following open questions for further study.

- Is there a polynomial-time exact algorithm for the minimum-area convex intersecting polygon of segments?
- Is there a polynomial-time exact algorithm for minimum-perimeter or minimum-area convex intersecting polygon of convex polygons, or are these problems NP-hard?
- Is there a polynomial-time approximation scheme for the minimum volume or minimum surface area convex intersecting polytope of convex polytopes in $\mathbb{R}^3$? Can we at least approximate the diameter of the optimum solution to these problems?

It would be especially interesting to see an NP-hardness proof for minimum volume or surface area convex intersecting set of convex objects in higher dimensions.

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## References

[1] Antonios Antoniadis, Krzysztof Fleszar, Ruben Hoeksma, and Kevin Schewior. A PTAS for Euclidean TSP with hyperplane neighborhoods. *ACM Trans. Algorithms*, 16(3):38:1–38:16, 2020. doi:10.1145/3383466.

[2] Keith Ball. Ellipsoids of maximal volume in convex bodies. *Geom. Dedicata*, 41(2):241–250, 1992. doi:10.1007/BF00182424.

[3] Saugata Basu, Richard Pollack, and Marie-Françoise Roy. On the combinatorial and algebraic complexity of quantifier elimination. *J. ACM*, 43(6):1002–1045, 1996. doi:10.1145/235809.235813.

[4] Bart Braden. The surveyor's area formula. *Coll. Math. J.*, 17(4):326–337, 1986. doi:10.1080/07468342.1986.11972974.

[5] Svante Carlsson, Håkan Jonsson, and Bengt J. Nilsson. Finding the shortest watchman route in a simple polygon. *Discret. Comput. Geom.*, 22(3):377–402, 1999. doi:10.1007/PL00009467.

[6] Moshe Dror, Alon Efrat, Anna Lubiw, and Joseph SB Mitchell. Touring a sequence of polygons. In Lawrence L. Larmore and Michel X. Goemans, editors, *STOC 2003: 35th Annual ACM Symposium on Theory of Computing*, pages 473–482, 2003. doi:10.1145/780542.780612.

[7] Adrian Dumitrescu. The traveling salesman problem for lines and rays in the plane. *Discret. Math. Alg. Appl.*, 4(4):1250044, 2012. doi:10.1142/S1793830912500449.

[8] Adrian Dumitrescu and Minghui Jiang. Minimum-perimeter intersecting polygons. *Algorithmica*, 63(3):602–615, 2012. doi:10.1007/s00453-011-9516-3.

[9] Sariel Har-Peled. *Geometric approximation algorithms*, volume 173 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2011. doi:10.1090/surv/173.

[10] Ray A. Jarvis. On the identification of the convex hull of a finite set of points in the plane. *Inf. Process. Lett.*, 2(1):18–21, 1973. doi:10.1016/0020-0190(73)90020-3.

[11] Ahmad Javad, Ali Mohades, Mansoor Davoodi, and Farnaz Sheikhi. Convex hull of imprecise points modeled by segments in the plane. In Jan Vahrenhold, editor, *EuroCG 2010: 26th European Workshop on Computational Geometry*, pages 193–196, 2010. URL: https://eurocg.org/2010/proceedings.pdf.

[12] Yiyang Jia and Bo Jiang. The minimum perimeter convex hull of a given set of disjoint segments. In *Recent Developments in Mechatronics and Intelligent Robotics*, pages 308–318, 2018. doi:10.1007/978-3-319-70990-1_46.

[13] Shunhua Jiang, Zhao Song, Omri Weinstein, and Hengjie Zhang. A faster algorithm for solving general LPs. In Samir Khuller and Virginia Vassilevska Williams, editors, *STOC 2021: 53rd Annual ACM Symposium on Theory of Computing*, pages 823–832. ACM, 2021. doi:10.1145/3406325.3451058.

[14] Fritz John. Extremum problems with inequalities as subsidiary conditions. In *Studies and Essays Presented to R. Courant on his 60th Birthday*, pages 187–204. Interscience Publishers, New York, 1948. doi:10.1007/978-3-0348-0439-4_9.

[15] Håkan Jonsson. The traveling salesman problem for lines in the plane. *Inf. Process. Lett.*, 82(3):137–142, 2002. doi:10.1016/S0020-0190(01)00259-9.

[16] Marc J. van Kreveld and Maarten Löffler. Approximating largest convex hulls for imprecise points. *J. Discrete Algorithms*, 6(4):583–594, 2008. doi:10.1016/j.jda.2008.04.002.

[17] Maarten Löffler and Marc J. van Kreveld. Largest and smallest convex hulls for imprecise points. *Algorithmica*, 56(2):235–269, 2010. doi:10.1007/s00453-008-9174-2.

[18] Nimrod Megiddo. Linear programming in linear time when the dimension is fixed. *J. ACM*, 31(1):114–127, 1984. doi:10.1145/2422.322418.

[19] Franco P. Preparata and Se June Hong. Convex hulls of finite sets of points in two and three dimensions. *Comm. ACM*, 20(2):87–93, 1977. doi:10.1145/359423.359430.

[20] Xuehou Tan. The touring rays and related problems. *Theoret. Comput. Sci.*, 866:43–55, 2021. doi:10.1016/J.TCS.2021.03.005.

[21] Csaba D. Tóth, Joseph O'Rourke, and Jacob E Goodman. *Handbook of Discrete and Computational Geometry*. CRC press, 2017. doi:10.1201/9781420035315.

## A Appendix: Counterexample to Lemma 6 of Tan [20]

In this section we give a counterexample to a lemma of Tan [20] that is used to establish his results on the TSPN problem for rays: given a set $\mathcal{R}$ of $n$ rays in the plane, compute the shortest closed curve (a tour) intersecting all rays in $\mathcal{R}$.

Tan [20] defines a tour to make a *crossing contact* with a ray $r$ (resp. a line $\ell$) if the tour has a non-empty intersection with both open half-planes defined by the supporting line of $r$ (resp. the line $\ell$).[3] Then, the concept of a *pseudo-touring rays route* is introduced, which is a tour starting and ending at some fixed point $s$ that is required to visit the supporting lines of all rays in $\mathcal{R}$, with the restriction that it must make a crossing contact with the supporting line of any ray that is not visited.

The shortest pseudo-tour for a starting point $s$ and ray set $\mathcal{R}$ is denoted by $\mathrm{POPT}_s(\mathcal{R})$, and the optimum tour visiting the rays and $s$ is denoted by $\mathrm{OPT}_s(\mathcal{R})$. We note that [20] shows that the optimal tour and pseudo-tour through $s$ are respectively unique (cf. Lemmas 4 and 5 in [20]). Now let $\mathcal{H}$ denote the convex hull of $\mathrm{POPT}_s(\mathcal{R})$ and the origins of the rays that are not visited by $\mathrm{POPT}_s(\mathcal{R})$. The result of Tan is based on the following lemma.

**Lemma 6 of [20].** *Assume that $R$ is a ray-segment such that it is not visited by $POPT_s(\mathcal{R})$ and its origin $p$ is on $\mathcal{H}$. Then, $OPT_s(\mathcal{R})$ makes a crossing contact with $r$.*

We prove the following lemma, which directly contradicts Lemma 6 in [20]:

**Lemma 27.** *There exists an input ray set $R$ such that the following hold: some ray $r$ is not visited by $POPT_s(\mathcal{R})$, the starting point of $r$ is on $\mathcal{H}$, and $OPT_s(\mathcal{R})$ does not make a crossing contact with $r$.*

*Proof.* We present an input instance consisting of a starting point $s = (0, -\epsilon)$, for some $\epsilon < 0.1$, and three rays defined as follows (see also Figure 12):

- $r_1$ has $y = \frac{4}{5}x + \frac{4}{5}$ as supporting line, $o_1 = (-1, 0)$ as starting point and the ray is pointing downwards.
- $r_2$ has $y = 1$ as supporting line, starting point $o_2 = (-1, 1)$ and the ray is pointing towards the right.
- $r_3$ has $y = 0$ as supporting line, its starting point is $o_3 = (2, 0)$ and the ray is pointing towards the right.

Let $T_p$ be a pseudo-tour going from $s$ to point $p_1 = (0, 1)$ and back to $s$. Note that $T_p$ is a feasible pseudo-tour starting and ending at $s$, since it reflects on $r_2$ and makes crossing contact with the supporting lines of $r_1$ and $r_3$. In particular, the line of $r_1$ intersects the line of $r_2$ at $p_2 = (\frac{1}{4}, 1)$, thus $T_p$ crosses the line of $r_1$ somewhere on $o_1p_2$. Furthermore, let $T$ be a tour starting at $s$, visiting $r_1$ at $o_1$, $r_2$ at point $p_3 = (\frac{1}{2}, 1)$ and $r_3$ at $o_3$ before returning

[3] We note that [20] actually defines a *crossing contact* in a more involved way, but for the purposes of our counterexample, the definitions are equivalent.

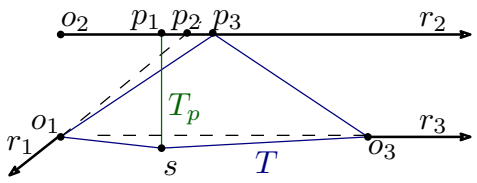



Figure 12: A counterexample to the statement of Lemma 6 from [20]. The dashed segments indicate the supporting lines of the rays $r_1$ and $r_3$.

to point $s$. This is a feasible tour starting and ending at $s$ since it visits all three rays of the input instance. See Figure 12 for an illustration of the input instance as well as $T_p$ and $T$.

We claim that $T_p$ and $T$ are an optimal pseudo-tour and tour respectively for the input instance. The lemma would then directly follow for $r = r_1$, since $r_1$ is not visited by $T_p$, its origin $o_1$ is on $\mathcal{H}$ which is defined by the vertices $o_1, p_1, o_3$ and $s$, and $T$ does not make a crossing contact with $r_1$ at $o_1$; it does what Tan [20] calls a bending contact. Since the optimal tour and pseudo-tour are unique the lemma then follows.

Regarding the optimality of $T_p$, note that any feasible pseudo-tour must either reflect on $r_2$ or make a crossing contact with the supporting line of $r_2$. Since the distance between $s$ and the supporting line of $r_2$ is $1+\epsilon$, any feasible pseudo-tour for the instance must therefore have a total length of at least $2+2\epsilon$. Since $T_p$ has a total length of $2+2\epsilon$ it must be optimal.

Regarding the optimality of $T$, we first observe that due to the fact that $r_1, r_2, r_3$ are pairwise disjoint and neither contains $s$, the optimal tour is a quadrilateral with one vertex at $s$, and one vertex on each of $r_1$, $r_2$ and $r_3$. We first show that $T$ is shortest among all tours that start from $s$, visit $r_1$ followed by $r_2$ and then $r_3$ before returning to $s$. We will then show that any tour visiting the rays in a different order must have a strictly higher cost.

By construction, the shortest line segment connecting $s$ and $r_1$ is $so_1$ and similarly the shortest possible line segment connecting $s$ and $r_3$ is $so_3$. Furthermore, the shortest possible path from $r_1$ to $r_3$ through $r_2$ is the one from $o_1$ to $p_3$ to $o_3$, where $p_3$ is defined as the point of $r_2$ such that $|o_1p_3| = |p_3o_3|$; see Figure 12. Therefore $T$ is the shortest possible tour that goes from $s$ to $r_1$ to $r_2$ and then $r_3$ before returning to $s$. Note that the length of $T$ is exactly $\sqrt{1+\epsilon^2} + 2\sqrt{1.5^2+1} + \sqrt{4+\epsilon^2}$ which is strictly less than 6.8 for any $\epsilon < 0.1$.

The only two other distinct possible orders of visitation are $s, r_2, r_1, r_3$ and then going back to $s$, and $s, r_1, r_3$ and $r_2$ before returning to $s$. The first one has a total length of at least $(1+\epsilon) + 1 + 3 + \sqrt{4+\epsilon^2}$ by taking the respective minimum distances between the objects. This total length is strictly greater than 7. In a similar way, the second one has a total length of at least $\sqrt{1+\epsilon^2} + 3 + 2\sqrt{2}$ which is strictly greater than 6.8. This concludes the proof. □